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HAS THE GREEN DEAL SPED UP COMPLIANCE?

Evidence from a Survival Analysis of Directives of the European Union

Author: Nicolò Marchini

Matriculation number: 33967A

Supervisor: Prof. Fabio Franchino

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To my grandma,

who, from the sky above,

I am sure has never looked away from me.

Abstract

The results show a significant change in the compliance dynamics of environmental directives linked to the European Green Deal. Directives adopted after the launch of the programme are transposed up to 30 times more slowly compared to those adopted before 2019. This slowdown reflects both the high politicisation of climate-related policies and the greater technical complexity that makes their implementation more demanding at the national level. Countries with an efficient bureaucratic apparatus and stable political systems transpose directives roughly twice as fast as Member States with weaker administrative structures and political instability. This confirms that administrative capacity and good governance are key to compliance: strong political will alone cannot compensate for weak institutions, but as will be discussed, also the opposite is true. This thesis analyses the transposition process of the European Green Deal directives, with the aim of understanding how institutional, political, administrative and socio-economic factors influence Member States' compliance and whether media coverage and public opinion also play a role. The research framework builds on the theories of the Management School and the Enforcement School but proposes a hybrid approach that interprets compliance as the result of the interaction between bureaucratic capacity and political pressure. Using an original dataset built from EUR-Lex and complemented with data from ParlGov, the Manifesto Project, Eurostat, Eurobarometer and the World Bank, the analysis applies Cox proportional hazard models and Kaplan–Meier survival curves to estimate what influence the directive transposition. The extension of the model to directives from labour sector, highlighted how the nature of policies inevitably shapes compliance behaviour. Environmental directives are more politicised and reflect dynamics of strategic confrontation, while labour directives remain anchored in more technical and procedural logics. In conclusion, the Green Deal represents a genuine test of the European Union's ability to translate ambitious objectives into concrete results.

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Introduction

The main question that this thesis aim to answer is: Has the Green Deal sped up compliance? As the empirical analysis will demonstrate, the answer is negative. The European Green Deal appears to have slowed down the transposition process, reflecting the increased policisation and technical complexity of its directives. To explain this answer, the thesis examines how institutional, political, administrative and socio-economic factors influence the speed at Member States transpose Green Deal directives. The European Union has long defined itself as a unique political system, capable of combining supranational governance with national sovereignty. Within this framework, Member States' compliance with EU law represents a crucial stage in the policy process, as it is the time when decisions taken at the EU level acquire practical effect through implementation within national legal frameworks. Understanding what shapes compliance with EU law and why some Member States perform better than others, is essential to assess the European Union's ability to achieve its ambitions. The European Green Deal is not merely an environmental strategy, but rather a comprehensive plan for economic and political transformation that seeks to redefine sustainability as both a driver of wealth creation and a tool for social justice. However, the size and ambition of this transition have increased its politicisation, generating both support and resistance across Member States. The coexistence of ambitious targets, multiple policy sectors and binding deadlines makes the Green Deal an ideal case for studying the mechanisms and dynamics underlying compliance within the European Union. To address this question, the thesis applies survival analysis models, specifically, Cox Hazard models and Kaplan-Meier curves to estimate how different variables affect the timing of directive transposition. By combining theoretical insights with quantitative analysis, this thesis contributes to the broader debate on compliance and European governance, showing that the Green Deal represents a concrete political laboratory to understand how institutional design and policy variation influence the effectiveness of European integration.

Chapter 1: Compliance in the European Governance

The first chapter provides the conceptual and theoretical foundations for understanding compliance within the governance system of the European Union. It builds on the material presented in Professor Franchino's lectures on EU governance, as well as on key academic contributions discussed throughout the course.

1.1 What is a directive and how it is approved

In the context of European Union Law, a directive is type of secondary law that set an objective that Members States must achieve but leave to them the decision of how to implement it[1]. Unlike regulations, directives must be implemented by all the Members States to enter in power, this happens by translating their content into national law. Directives such as those included in the European Green Deal, which cover areas like environmental policy, are adopted through the Ordinary Legislative Procedure involving the European Commission, the European Parliament, and the Council of the European Union. The process starts when the European Commission makes a proposal and sends it to both the Parliament and the Council. Then, the Parliament, in first reading, decides whether to accept it or amend. After that, the Council does the same by accepting the Parliament's version or make its own amendments. The text then goes back to the Parliament for a second reading, where it can again be accepted, rejected, or amended. Finally, the Council reviews it one more time and decides whether to agree with the Parliament's version. If the two sides still do not agree, a special committee is formed to find a compromise. If a compromise is found, the text is sent for a third reading in both the Parliament and the Council. Only if both institutions approve the final version the directive becomes law. If either of them rejects it, the proposal fails. The voting system within the council is the qualified majority, following the rule of the Treaty of Lisbon[2]. This mechanism allows for the adoption of certain directives even if some

[1] Article 288 TFEU.
[2] Article 238 TFEU.

governments oppose them and leave to the Commission a greater discretion, since the pivotal coalition is more likely to share its preferences (Franchino 2004). Once adopted, a directive requires the Member States to translate it into their respectively national legal systems within a given deadline (usually from 18 to 24 months). Transposition represents the first step towards compliance. However, since EU law leaves discretion to Member States in deciding how to implement directives, this inevitably leads to differences in timing and approaches to implementation.

1.2 The Compliance in the European Governance Context

In the context of the European Union, the term "compliance" refers to the respect of European law by the Member States and represents a crucial component of multilevel governance. Rather than being just a formal step, it is a complex administrative and political process that involves multiple actors a different level. The logic of multilevel governance implies constant interaction between European institution, national governments, public administrations and in some cases even sub-national authorities. The implementation of EU law does not occur automatically; each directive must be adapted to the legal system of every Member State. As already explained, this process varies in timing and modality according to each country discretion. Several studies (Tallberg 2002; Dimitrova et all. 2009) showed that administrative capacity, meant as the ensemble of technical, human and originative resources at disposal of public administrations at disposal of public administration is a key determinant to compliance. States with good bureaucratic institutions and high level or government effectiveness tends to transpose faster the EU directives. In contrast, Member States with weak or fragmented administrations may encounter some difficulties even in presence of political will. Some studies (Fjelstul and Carrubba 2018; König and Mäder 2014) underlines how political will also plays a crucial role. Compliance is never a neutral or purely administrative process; it is shaped by the political priorities and strategic calculations of governments. Cabinets may delay, obstruct, or manipulate compliance for strategic reasons for instance, to signal opposition to certain EU policies or to avoid domestic political costs. As shown by König and Mäder's (2014) strategic model of compliance, the process can be imagined as a game theory interaction between the Member State and the European Commission, where both actors make rational

decisions about whether to comply or enforce based on costs and benefits. These insights reflect the broader debate of why states comply with EU law. Two main explanations emerge: one highlights administrative capacity and problem-solving, while the other points to political resistance.

1.3 Theories of Compliance

In the context of multilevel governance of the European Union, compliance represents a crucial moment of the integration process as it not a merely technical or juridical step, but it is the result of complex balance between state capacity, political will and institutional commitment. The literature on compliance has developed two main theories to explain the causes of noncompliance: the Management School and the Enforcement School, as explained in Tsebelis (2002). The Enforcement school sees compliance through political economy and collective action theory. It describes states as rational actors who evaluate the cost and benefits obeying EU Law. Noncompliance occurs when the benefits from delaying or avoiding implementation are higher than the cost of being sanctioned. In this perspectives, a state may prefer to not comply when doing politically or economically costly. The European Commission, acting as the "guardian of the Treaties", in such cases, enforces compliance primarily through monitoring and infringement procedure, which may lead to sanctions by the European Court of Justice (ECJ). By contrast, the management school offers a more optimistic view, arguing that noncompliance is usually unintentional and often results from capacity limitations or ambiguity in the rules, rather than from deliberate resistance. The remedy to these situations involves capacity building, rule interpretation and transparency. These are implemented through tools such as administrative cooperation programs, interpretative guidelines and EU funds aimed to support national administrations. These two theories are not mutually exclusive but rather complementary within the EU governance system. Mbaye (2001), try to explain why Member States differs in their capacity of comply with directives. She argues that the causes of noncompliance are not idiosyncratic but rather structural and therefore comparable at cross-country level. According to her analysis, noncompliance may come from administrative constraints, such as inefficient bureaucracies due to limited size, lack of resources or corruption. But this is not the

only reason, as noncompliance can alternatively be seen as the result from strategic choices, when governments deliberately decide to not fully implement a directive that does not match with their preferences in the European Council. As explained before, when a directive reaches the Council for the first or second reading, it is approved through the qualified majority voting (QMV) system. Mbaye (2001) argues that QMV can increase the number of infringements, as it removes the countries' veto power, allowing directives to be adopted even against some states' preferences. However, she also suggests that over time, institutional learning and adaptation help to reduce noncompliance. Moreover, she also suggests that federal states or strongly regionalised systems shows more infraction, since the centrale government has less control over the subnational units and that cabinets with a weal support towards European integration implements directives with more difficult to avoid internal political costs. On these contribution, König and Mäder (2014), as saw, introduce a game theory model of compliance that inspect the relationship between the European Commission and the Member States.

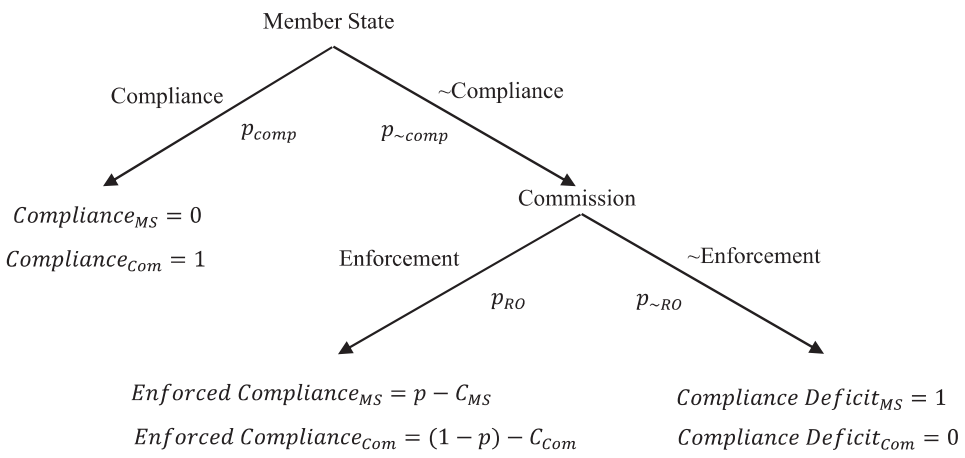

Figure 1: Strategic Model of Compliance and Enforcement in the European Union[3]

Where:

[3] The figure is taken from König and Mäder (2014) and explained on page 251.

C_{MS} is the cost for Member State.

C_{Com} is the cost for Commission.

p is the probability of compliance[4].

Figure 1, shows the model adopted by the authors that displays the strategic behavior by the Member States that decide whether to comply, anticipating whether the Commission will take an enforcement action. On its side, the European Commission choose whether to start an infringement procedure based on the probability of success and the cost of sanctioning. In their analysis, Thomas König and Lars Mäder argue that the Commission tends to not proceed by initiating sanctions when the chances of winning are low or when the expected costs of litigation are too high. Conversely, Member States are more likely to avoid compliance when they strongly disagree with a directive or when divergent preferences among governments weaken the Commission's position. These explanations show that compliance is not a single process but rather is the result of mutual expectation between the Commission and the Member States. While the Commission acts respect to its limitations, national governments balance administrative capacity with political interests. Compliance is therefore a dynamic process shaped by institutional design, state bureaucratic capacity and political will.

Chapter 2: The European Green Deal

2.1 Defining the European Green Deal

The European Green Deal is probably the most ambitious policy strategy adopted by the European Commission. The Green Deal aims to represents the strategy of the Commission to make Europe the first climate-neutral continent by 2050 and its part of the strategy to implement the United's Nation 2030 agenda. The goals set

[4] These definitions and the logic of strategic interaction are based on Franchino (Lecture 8, Governance of the European Union), which draws on König and Mäder (2014) to explain compliance as a probabilistic and cost-sensitive process within EU multilevel governance.

out by the Green Deal are various and cover diverse topics. It aims to protect and recover biodiversity, promote circular economy, favor a just and inclusive transition and strength citizens' trust and participation. Furthermore, the Green Deal presents as both a policy framework and investment plan that set to brings opportunities and stimulate the development of lead markets for climate neutral and circular products. In short, the Green Deal turns the EU's environmental ambition into an economic transformation plan, a strategy for sustainable and inclusive growth that aims to make Europe a global leader in climate action and green innovation[5]. The Green Deal is funded through a financial plan amounting to €1,000 billion over the next decade. The resources come mainly from three sources. First, the EU budget, which allocates around €503 billion. Second, the InvestEU programme, which attracts public and private investments, generating approximately €279 billion. Third, the Just Transition Mechanism, which provides €100 billion to support workers and regions most affected by the green transition. Together, these instruments combine public funds, financial guarantees and co-financing mechanisms to sustain the economic transformation proposed by the Green Deal[6].

2.2 The Green Deal from a Political Point of View

From a political point of view, the European Green Deal represents the will of the European Union to propose a new initiative: the green growth. In politics terms this mean redefine the European economic-social contract by reallocating resources and priorities from traditional sectors towards the green transition. This transformation creates winners and losers, as the redistribution some sectors and region will have benefit while other will inevitably loose from it. Some European governments, especially those from countries that strongly depend on coal or gas for their energy supply, criticised the reform as to their eyes the Green Deal represents an inference of the Commission in their national affairs. Particularly, the country of the Visegrád group (Poland, Hungary, Czechia and Slovakia) expressed a clear opposition

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[5] European Commission. The European Green Deal, COM(2019) 640 final, Brussels, 11 December 2019.

[6] Europe Direct Trapani Sicilia, The European Green Deal, Next Generation EU and the PNRR: Acting Today and Shaping Tomorrow (Trapani: Consorzio Universitario della Provincia di Trapani and European Commission Representation in Italy, 2021)

towards the green transition. In a joint declaration, the prime ministers of these countries have underlined the importance of protecting energy security and accessibility, highlighting the significant differences in income level across the EU and the necessity for huge investments that middle-low-income countries will face (Riepl and Zavarská 2022). From the point of view of the European parties, the Green Deal has triggered the phenomenon of "green backslash"[7]. The most sceptics parties over the Green Deal are right-wing populist, which follow the agenda of the elitist imposition by Brussell that threats national jobs, sovereignty and energy security. According to Thalberg et all. (2025), the rise of right-wing populism in Europe has made the Green Deal more politicised. These parties use the green transition as a symbol to oppose the European Union and its policies, which they say threat the European citizens. In Germany, the AfD turned the energy transition into a fight against its social costs. In Italy, Matteo Salvini's League and Giorgia Meloni's Brothers of Italy used measures such as Milan's "Area B and C" to argue that climate policies are unfair. In Poland, the Law and Justice Party (PiS) openly clashed with the European Commission, accusing it of threatening national energy security. Overall, Thalberg et. al (2025) shows that the Green Deal has become an identity issue for nationalist parties, reshaping political conflict inside and outside the European Parliament. By contrast, some studies also highlight the increased sustain to green parties across Europe. In fact, the support over the Green Parties in Western Europe, has constantly increased over the last decades, especially among the younger generations (Lichtin et al. 2023). The study shows that with each new electoral turn; voters are more likely to support Green parties compared to previous one. The authors argue that this trend comes from a broader shift in social values, as generations born after the 1960s and 1970s, raised in contexts of greater economic prosperity and higher education, tend to give more importance on post-materialist values such as environmental protection. In this sense, the expansion of the "green vote" represents the progressive side of the Green Deal's politicisation. While right-wing parties react with the so-called "green backlash", younger

[7] Elisabetta Cornago, "How to minimise the 'greenlash'," Insight, Centre for European Reform, 18 December 2023, https://www.cer.eu/insights/how-minimise-greenlash.

generations and urban, educated electorates push in the opposite direction, demanding more ambitious climate policies.

2.3 The Green Deal as a Legal and Institutional Framework for Compliance Analysis

From juridic and institutional point of view, the European Green Deal is not a single act, but rather it is policy framework made of an ensemble of Regulations, Communications, Directives and Strategies. It was presented by the Von der Leyen Commission I on December 11th 2019, as the strategy plan of the European Union towards the climate neutrality by 2050 and as commitment to the Onu 2030 agenda[8]. Despite being composed of many instruments, the Green Deal operates mainly with Regulations and Directives. Regulations, such as the European Climate Law, are directly applicable and binding on all Member States. They establish common targets, such as the reduction of at least 55% of CO_2 emissions by 2030 compared to 1990 levels. As previously explained, directives, for instance, Directive (EU) 2023/1791 on energy efficiency, set binding objectives but leave Member States discretion over the implementation, while imposing a specific transposition deadline. Directives are therefore central to the study of compliance, as the obligation to transpose them into national legal frameworks highlights the differences among Member States in terms of administrative capacity, efficiency, and coordination. The strong politicisation of the European Green Deal, along with its broad scope and clear deadlines, makes it a perfect case for studying how quickly Member States comply and what factors influence this process. Furthermore, the Green Deals represents an environment to test both the theories of compliance (management and enforcement school). At the same time, the need to meet precise deadlines for the implementation of directives makes it possible to assess how the speed of transposition affects the overall effectiveness of policies and the ability to achieve climate targets within the established timeframe. In this context, this thesis examines whether the European Green Deal has accelerated compliance with EU directives, analysing how political, institutional and administrative factors

[8] European Commission, The European Green Deal, COM(2019) 640 final.

influence the speed of transposition and ultimately, the effectiveness of Green Deal policy implementation.

Chapter 3: Constructing the Dataset and Designing the Empirical Analysis

This chapter explains how the dataset was built and how the empirical analysis was designed. The goal of this thesis was not only to study the transposition of Green Deal directives, but also to develop a model that could be reused for other policy areas and research on EU compliance. By combining institutional, political, administrative, media and public opinion data, the chapter provides both an original dataset and a method that can be applied beyond the Green Deal to better understand how and why Member States comply with EU law. The full list of data sources is provided in Appendix C of this thesis.

3.1 Research Strategy and Identification of Green Deal Directives

The empirical analysis of this project represents perhaps its most challenging but also its most interesting part. Given the decision to focus on the compliance of the European Green Deal, due to its nature and its novelty, it was clear that finding a ready to use dataset would be almost impossible. Nevertheless, this topic was chosen precisely for the challenge of construing the dataset from raw data and the opportunity to design the analysis according to personal criteria. This approach made possible to achieve a greater precision in identifying the factors that truly matters about the transposition of the European Green Deal's directives. Furthermore, this process ensured greater transparency in the data and provided an original contribution, since the dataset itself constitutes a resource that can be reused, updated and even extended in future research. The first step to build the dataset, was of course finding information regarding the transposition of the European Green Deal's directives. If on one hand, it is possible to find all the

information needed of a directive from the EUR-Lex platform[9], on the other hand, two significant problems emerged. First, as explained in the previous chapters, the Green Deal is not a single law but rather a policy package composes of many legal acts in various forms, including regulations, decision, directives, and other non-legislative measures. However, to study for the compliance of the Green Deal, the focus had to be put exclusively on the directives as they represent the only type of legal act that creates a "time to event" compliance dynamic through national transposition. Second, despite all efforts put in the research, no official list of "European Green Deal measures" could be found and EUR-Lex does not provide any label identifying which directives are part of the Green Deal. Fortunately, it was possible to find two official sources that provided this information. The first was the Fit for 55 package, a set of legislative proposals aimed at ensuring that the EU will reduce its net greenhouse gas emissions by at least 55% by 2030."[10]. Although the Fit for 55 package is not identical to the Green Deal, is a central component of it[11]. From the Fit for 55, it was possible to identify many of the directives included later in the dataset. The second source found, was the European Parliament's Legislative Train Schedule, an online tool that tracks both the legislative and non-legislative files of the European Union. Each "train" groups the related files under a thematic umbrella, such as the European Green Deal[12] in this very case. The platform indicates also the procedural status of each file (e.g. announced, tabled, close to adoption, adopted), which made possible to track which measures had already been adopted. Based on these findings, the dataset includes only newly adopted directives but also those that amend pre-existing legislation, since they create an obligation for Member States to transpose national

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[9] Publications Office of the European Union, About EUR-Lex, EUR-Lex, accessed October 2, 2025, https://eur-lex.europa.eu/content/welcome/about.html.
[10] European Commission, Delivering the Green Deal, Fit for 55: Delivering on the Proposals, accessed October 2, 2025, https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/european-green-deal/delivering-european-green-deal/fit-55-delivering-proposals_en.
[11] European Commission, Delivering the European Green Deal, accessed October 3, 2025, https://commission.europa.eu/strategy-and-policy/priorities-2019-2024/european-green-deal/delivering-european-green-deal_en.
[12] European Parliament, The European Green Deal, Legislative Train Schedule, accessed October 3, 2025, https://www.europarl.europa.eu/legislative-train/theme-a-european-green-deal/file-european-green-deal.

implementing measures. Because the Green Deal, at the time the analysis was conducted was only six years old, this made it possible to enlarger the sample of directives. The full list of directives is provided in Appendix D of this thesis.

3.2 Data Collection on Directives and National Transposition

After that all the list of directives part of the European Green Deal were found, the only remaining thing to go on EUR-Lex and download all the information needed. Unfortunately, EUR-Lex was not originally created as a portal for citizens or researchers. Its roots go back to CELEX (Communitatis Europae Lex), developed internally in the Legal Service of the European Commission in the late 1960s and early 1970s as a professional tool for managing the growing number of Community laws[13]. In 1994, CELEX was first opened to the public, but it remained a service mainly for professionals[14]. Between 2001 and 2004, CELEX and EUR-Lex were progressively merged, and by 2004 EUR-Lex had become the single, free access point to EU law, fully incorporating the CELEX database[15]. This is the reason why, no option to download a comprehensive dataset for the directives was available on the website and is when the second problem emerged too. The transposition is a process where a Member States of the European Union implement a directive into their national legislation system. Each directive describes a set of goals that the Member States must achieve, but it does not specify how, so it is up to national authorities to decide. Some countries adopt only one or a few national measures to transpose a directive, while others need many legislative or administrative acts. To make practical example, the "Directive (EU) 2018/2001 of the European Parliament and of the Council of 11 December 2018 on the promotion of the use of energy from renewable sources" has a total of transposition acts that surpass the thousand and countries like Austria and Hungary needed more than one hundred of acts to transpose this directive[16]. Since it was not possible neither to download these data

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[13] Publications Office of the European Union, 25 Years of European Law Online (Luxembourg: Publications Office of the European Union, 2006), 16.
[14] Ibid., 9.
[15] Ibid., 18.
[16] EUR-Lex, Directive (EU) 2018/2001: National Transposition Measures, accessed October 3, 2025, https://eur-lex.europa.eu/legal-content/EN/NIM/?uri=CELEX:32018L2001.

directly nor copy and paste them in Excell in a tabular format, the only apparent was to record manually one by one. However, this would been extremely time-consuming and would have increased the risk of error. This scenario required the acquisition of specific technical skills. These were developed through a combination of both self-study and formal training. On the one hand, standard references such as VanderPlas (2016) and Downey (2015) provided a solid foundation in data manipulation, automation, and reproducibility. On the other hand, the course of Data Science and Public Policy, offered practical training in applying fata science tools to policy-oriented research. Using Visual Studio Code together with Jupyter Notebook, it was developed a python script that, given the URL of an EUR-Lex/NIM page, automatically collected information such as the CELEX code, the country, the measure title, the transposition deadline, and the publication date of each national implementing measure. Thanks to the script, the extracted information was automatically organised into CSV files, with one file for each directive. At this point, another limitation of the EUR-Lex platform emerged. This database does not always show the effective date of entry in force, the deadline, or the date adoption by the Member States. In many cases, this information was missing and replaced by the placeholder "1/1/1001". While it was possible to retrieve the information concerning the directives themselves by consulting the "ALL" page or reading the official text, finding details such as the publication date for each act of transposition by the Member States was longer and harder. This task took most of the time of the data collection phase, since hundreds of missing dates had to be searched and recorded manually[17]. After this extensive work, the information about the directives where complete (even though it was not possible to retrieve all the missing dates for the transposition acts) and thanks to another a Python script, all the CSVs were merged to create the first version of the dataset that would later serve as the basis for the survival analysis.

[17] The publication dates of national transposition measures were verified, whenever possible, against the Official Journals of the respective Member States. Given the large number of sources, it is not feasible to list each individual reference in this thesis, although the specific journals consulted were recorded during the data collection process.

3.3 Theoretical Framework and Variable Selection

Beyond the analysis of the two main schools of compliance, it was essential to draw on a wider range of studies to identify potential factors explaining the timing of transposition. The aim was not only to account for compliance with Green Deal directives but also to build a framework that could be reused and adapted in future research on directive transposition. It was therefore crucial to find proper variables that would design the analysis. Very important was the process of selection, since these variables had to be relevant and explanatory first of the pace at a directive is transposed and second, they had to be related to the climate change and Green Deal topic. The literature consulted before starting this thesis was in great majority of studies on compliance, but each of them analyse different factors. Some studies focus their research on structural and procedural characteristics of EU directives, noting how aspects such as legal complexity, available time until the deadline, and whether a directive is new or amending influence national compliance delays (Mastenbroek 2003). Steunenberg and Rhinard (2010) explore different directive-level characteristics, such as whether a directive is new or modifies an existing one, its normative complexity, and the time available before the transposition deadline, along with the intensity of monitoring by the Commission. The authors also examine national conditions, including public support for the EU, experience with transposition, the number of actors involved, and timing relative to elections. Finally, they consider process-related factors, such as the number of veto players involved, the nature of legal instruments deployed, procedural hurdles, and the passage of time since directive adoption. Other characteristics were identified in further studies, such as the types of legal instruments used, the structure and coordination of ministries, the number of ministries involved, and the influence of both formal and de facto veto players, as well as the role of monitoring by the European Commission in shaping implementation (Spendzharova and Versluis, 2013). While Zhelyazkova and Torenvlied (2009), examines policy conflict in the European Council, issue salience among both governments and the public, and the presence of Green parties in national governments. Since the topic of the European Green Deal is something widely debated in the society, it was important to include not only the characteristics of the directives, such as the publication date of the

transpositions or the deadlines, but also other factors that could shape the national context. These include state level and political of the governments responsible for the transposition, as well the salience of climate change and the Green Deal in the media and finally the public sentiment towards these topics. The purpose was to assess whether the birth of the Green Deal increased the pace of transposition for climate policy directives, and to identify also what are the element that influence how fast a directive is implemented. Before explaining how these variables were found and collect, it is important to highlight that on the initial dataset, obtained by merging the information regarding the Green Deal directives downloaded from EUR-Lex, some variables, always utilising a Python script, were created. Drawing inspiration from the study Steunenberg and Rhinard (2010), the time interval between each directive's transposition deadline and the publication date of the corresponding national measures was calculated and stored as a variable. The value was recorded as negative if the transposition occurred before the deadline and positive if after. The time difference was calculated in days but also registered in terms of weeks and months. This was a crucial step, as the variable stored in terms of months later served as the basis for the dependent variable in the survival analysis.

3.4 Media Salience and Public Attention

Following the consultation of the papers cited above, it was decided to include variables not only related to the characteristics of the directives, but also to broader contextual factors. These included the institutional characteristics of the Member States, the ideological orientation of governing cabinets, not only along the traditional left–right spectrum, but also in relation to climate and environmental policy, the media salience of climate issues and the Green Deal, as well as public opinion towards these topics. First, were collected the data on the media salience among the Member States for each year. This was a very difficult collection, as is almost impossible to assess how much space these arguments had among the media in each EU Member State. What was done, was to shape this variable thanks to two tools that are the Media Salience website and Google Trends that, in different ways, take note of how trending is a topic online. Google Trends is well known and allows people to inspect the popularity of a search query or of a topic over time and across

different countries but also region or cities[18]. Media Cloud is research project developed at Harvard, the University of Massachusetts Amherst, and Northeastern University. It is an open-source tool, that allows to conduct quantitative and qualitative analyses of online media content. Using the Media Cloud platform, is possible to investigate how many sources cover topics, the language used in each of them, and how they are spread across time and among countries[19]. Unfortunately, Media Cloud has a limitation as its databank at this moment covers only the years from 2014 upward, while Google Trends ha registered all the quires done since Google launch. The Media Cloud databank was used to create a Media Salience indicator, which reflects the visibility of climate-related topics in online news media. For each country, the number of climate-related articles published each year was then divided by the total number of articles over the same period. The data were after processed using a Python script to produce the annual measure of media salience related to climate change for each country[20]. In parallel, Google Trends was used as source for the "Relative Climate Salience" indicator. If the problem for the Media Cloud platform is that no data is available before 2014, Google Trends shows the records only in relative terms, meaning that search interest is normalised on a scale from 0 to 100 within each country and topic. This makes direct cross-country or cross-topic comparisons problematic, as a value of 100 does not indicate the same absolute volume of searches across countries. Instead, it reflects the peak interest within that specific national context over the selected period. To address this problem, the Relative Climate Salience indicator was calculated by dividing the average annual interest for the topics "climate" and "green deal" by that for "economy". This calculation allowed to have a measure more stable and easier to be interpreted. While it may appear as an imperfect approach, measuring of how salient climate issues were compared to economic concerns within each Member State, made possible to compare how important climate issue were over time and

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[18] Google News Initiative, "Google Trends: See What's Trending Across Google Search, Google News and YouTube," accessed October 8, 2025, https://newsinitiative.withgoogle.com/en-gb/resources/trainings/google-trends-lesson/.
[19] Media Cloud, "Media Cloud Project Overview," Berkman Klein Center for Internet & Society, Harvard University, accessed October 8, 2025, https://cyber.harvard.edu/research/mediacloud.
[20] For the year 2025, only the first six months were considered to maintain consistency with the study's time frame.

across countries. With a different approach, the two variables measure how much the topic "climate" is spread across online media and among public. The values were recorded from 2014 to June 2025 for Media Cloud (data collection took place in July 2025) and from 2004 to June 2025 for Google Trends. Finally, the two variables were merged with the initial dataset by country name and by year, which was derived from the publication date of each national measure, so that we could match the corresponding media salience with the the measure was approved.

3.5 Government Ideology and Political Determinants

To explain how political factors may influence the transposition of directives, this section builds on the Enforcement School of compliance. To account for these political factors, data were first drawn from the ParlGov Database, an opensource database that provides information on elections, parties and governments for nearly all the OECD countries (including all the EU Member States) from the Post-war period to the present[21]. The data downloaded from ParlGoV were organised into three datasets: "election", "party" and "cabinet". Each of them containing information potentially relevant to explain the dynamic of directive transposition. The "election" table contained data on each party's vote share and the number of seats obtained in the parliament in each election. The "party" table provided data in the ideological position for each party across several dimensions: *Left-Right*, *State-Market*, *Liberty-Authority* and *Pro-Anti EU*. The positions were recorded on a scale from 0 to 10, where lower values indicate a strong alignment with the first position in the dimension label (e.g., a value of 0 on the State–Market scale denotes strong support for state-led economic policy). The methodology for how these variables were coded is described into the ParlGoV Codebook, which was consulted to correctly interpret the values[22]. The last table "cabinet" listed the composition of each government, including the parties involved and the start date of each cabinet. Since this relevant information were spread across different tables, it was not ready to be used into the analysis. Finally, thanks to another Python script, these tables

[21] ParlGov, About ParlGov Project, accessed October 9, 2025, https://parlgov.org/about/.
[22] ParlGov, ParlGov Codebook (version October 2023), accessed October 9, 2025, https://parlgov.org/about/.

were merged into a unified table thanks to the unique identifiers available for each party and election. The result was a clear dataset that for each cabinet, included all the parties in the coalition, the cabinet's start date, each party's strength (in both vote share and chamber seats) and their ideological position on the four dimensions mentioned before. To include variables that were explanatory of the environmental and climate position of national governments, data from the Manifesto project were used. The Manifesto project is a famous and widely used database that analyse and position the parties' position by inspecting their manifestos[23]. This dataset made possible to have a measure on the position that each party has on different topics such as, in this case, on policy regarding the environment and the climate change. To pursuit the topic of this thesis, it was necessary to focus on specific variables that were recording the position of each party on important environmental or climate issue. The variables chosen were *Environmental Protection* that captures the position on protecting the natural environment, *Climate Energy* that account for broader energy policy and its connection to climate change, *Renewables* that measures how much the party promotes renewable energy sources and *Sustainable Development* which shows how much a party emphasizes development that balances economic growth with environmental[24].

Thanks to these variables, it was possible to have a detailed picture of how much a cabinet is environmentally oriented and later to discover if having a government pro to this kind of policies would be a synonym of faster transposition. These data were then merged with the ParlGov dataset using unique identifier shared by both the datasets. Before merging the newly created dataset containing all the information about the cabinets with the one that already included data on directives and media salience, two additional steps were necessary. The first was essential as the ParlGoV dataset was not update to 2025. The latest version available at the time of data collection was from 2024 and cabinet data did not go beyond 2023 and in

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[23] Manifesto Project, Manifesto Project Database: Project & Coding Instructions, accessed October 9, 2025, https://manifesto-project.wzb.eu/information/documents/information.
[24] The name of the variables are taken from the Manifesto Project's official codebook. Manifesto Project. Manifesto Project Dataset Codebook, version MPDS2025a. Berlin: WZB Berlin Social Science Center, 2025.

some cases, were only up to 2022 even though several countries had experienced one or more cabinet changes in the meantime. Therefore, it was necessary to manually record all the change of cabinet happened from 2023 to June 2025 across all the EU Member States and then assign to each new cabinet the relative weight of each party in the coalition (and all the elections hold too), along with their positions using data form the ParlGoV and Manifesto datasets. The second step was to reshape the cabinet dataset. Up to that point, each cabinet was recorded across multiple rows (one for each party in the coalition) and each of them include the party's vote share, the number of parliamentary seats and the ideological positions. To have the ideological orientation for each cabinet, it was necessary to aggregate all the information of the parties. This was done by calculating the weighted average of the ideological variables of the coalition parties using each party's share of vote as weighted factor. After this process, each cabinet was assigned with the overall ideological profile (still expressed across the variables of DataGoV and Manifesto) reflecting the relative strength and the political orientation of its coalition.

3.6 Climate Concern in Public Opinion

As climate change has become a highly politicised topic in Europe, with strong public debate and wide media coverage, it was important to consider both party positions and citizens' views. Public opinion can push governments to act faster on Green Deal directives or instead make them more cautious when the costs of compliance seem too high. To assess how public opinion may shape the speed of a Green Deal directive; data from various waves of the Eurobarometer survey were used. Thanks to the GESIS data archive, one of the most important research infrastructure institutions in Europe [25]that hosts and archives the official Eurobarometer datasets[26], it was possible to relies on several surveys that offered a valuable source of information on citizens' perception of climate change. Covering the period of the analysis (from 2004 to 2025) was not easy at all, as first not every

[25] GESIS – Leibniz Institute for the Social Sciences. "About Us." GESIS.org. Accessed October 10, 2025. https://www.gesis.org/en/institute/about-us.
[26] GESIS – Leibniz Institute for the Social Sciences. "Eurobarometer Data Service – About." GESIS.org. Accessed October 10, 2025. https://www.gesis.org/en/eurobarometer-data-service/about.

year the Eurobarometer conducts surveys regarding the climate change topic and second, the survey always differs across them. The challenge was to find a way for harmonise all the surveys and build a consistent and comparable variable across time and countries and second find a method that would allow to cover all the years for the period of the analysis. Before, explaining how this issues were faced and solved, it is very important to explain why for the Eurobarometer was chosen only the topic "climate change" and the reason is because of surveys that account for the opinions towards such initiative of the European Union don't exist, only one exists but it was impossible be harmonised as it differs a lot from the other that were chosen. For the study period covered by this thesis, twelve Eurobarometer surveys related to the topic of climate change were identified. Although the wording of the questions between theme, they included at least one item indicating if climate change was considered a major concern, priority or issue. This question typically asked how serious the problem of climate change is perceived, which global or national issue are most important or what EU's policy main priority should be. To build a unified variable, this time, it was used Stata as all the datasets of the Eurobarometer downloaded from GESIS were in dta format. The variable in question was built by calculating the share if respondents in each country and year who explicit mentioned climate change in response to the survey' question. The surveys used included EB 68.2 in 2007; EB 71.1 in 2009; EB 75.4 in 2011; EB 80.2 in 2013; EB 83.4 in 2015; EB 88.1 in 2017; EB 92.4 in 2019; EB 95.1 in 2021; and EB 97.3 in 2022. For the year 2020 the values were taken from the Harmonised Eurobarometer Dataset 2004–2021[27] using the variable *issue_14* which grouped the answer to the question related to climate change. Despite the title, the Harmonised Eurobarometer Dataset 2004–2021 recorded this variable only for the years 2019, 2020 and 2021. Therefore, it was not possible to use this dataset for the whole period covered. After creating one harmonised dataset per year, the data were

[27] The Harmonised Eurobarometer 2004–2021 is a consolidated dataset covering 35 Standard Eurobarometer waves from 2004 to spring 2021. It standardizes over 140 variables related to EU policy attitudes, identity, institutions, and partisanship, using data published by GESIS. Example Stata do-files are provided to support replication. GESIS – Leibniz Institute for the Social Sciences. Harmonised Trend File Eurobarometer 2004–2021 (Version 3.0.0). 2023. https://search.gesis.org/research_data/SDN-10.7802-2539.

exported as separate CSV files and then merged them into a single dataset covering the period from 2007 to 2022. The result was a cross-country panel of climate risk perception over time but despite having harmonised the surveys, a limitation was still present. The years before 2007 and after 2025 were not covered because, as mentioned earlier the Eurobarometer surveys addressing climate change are not conducted annually. The second challenge was to identify a logical method for filling the missing years, otherwise it would have been impossible to include the public opinion in the analysis without dropping too many observations, especially as many of them correspond to transpositions that occurred after December 11th, 2019, the official launch date of the European Green Deal[28]. To solve this issue of missing values in the public opinion variable for the years without a Eurobarometer survey, while merging this data with the main dataset, a linear interpolation was applied within. Using this technique was possible to estimate for each state the values for the missing years based on existing data points. This approach allowed the construction of a continuous yearly time series of climate risk perception across all countries. However, this was not possible for the years before 2007 as the interpolation was conducted forward. In these cases, the values were given by assigning to each country the same value recorded in 2007 (the first survey that address the climate change issue). Linear interpolation was preferred over more complex imputation techniques because of its transparency and due to the relatively regular distance and stability of the data in the Eurobarometer waves. This ensured that the dataset could cover the entire period interested by the analysis while maintaining its consistency.

3.7 Data for Governance and Public administration performance

To analyse the quality of governance and the efficiency of public administrations across EU Member States, this section draws on the Management School of compliance. Data were collected from the Worldwide Governance Indicators (WGI), which is a global compilation of data collected by the World Bank that captures the perception of household, business and citizens towards the quality of

[28] European Commission, The European Green Deal, COM(2019) 640 final.

governance[29]. The WGI provides six indicators, each one capturing a key dimension of governance performance. These include *Voice and Accountability*, which reflects citizens' ability to participate in selecting their government and the extent of freedoms of expression, association and the media; *Political Stability and Absence of Violence/Terrorism* (from now in advance just *Political Stability*), which measures the perceived likelihood of pollical instability and/or politically motivated violence; *Government Effectiveness*, which captures perceptions of the quality of public services and of policy formulation and implementation; *Regulatory Quality*, assessing the government's ability to design and enforce policies and regulations; *Rule of Law,* which measures how much people trust and follow the laws and *Control of Corruption*, which reflects how much public power is seen as being used for personal gain[30]. Each of these indicators is scaled between approximately -2.5 (weak governance) and +2.5 (strong governance)[31] and is based on data from over 30 different sources, such as international organisation, NGOs, think tank and private firms. The data, captured through surveys, assessment and pools, are standardised and aggregated using a statistical technique called "Unobserved Components Model", which makes possible to combine several indicators and obtain the six indicators explained above[32]. For this analysis, were considered only the 27 EU Member States and the years from 20004 to 2023. Since the original dataset did not include values for 2024 and 2025, two years relevant to the transposition of several Green Deal directives, through a python script, I created these years by using data from 2023, using the same approach of the Eurobarometer data explained in the previous paragraph. In parallel, GDP per capita (in current euros) was collected from the Eurostat platform[33]. This variable represents an

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[29] Daniel Kaufmann and Aart Kraay, The Worldwide Governance Indicators: Methodology and 2024 Update, Policy Research Working Paper No. 10952 (Washington, D.C.: World Bank Group, 2024.
[30] World Bank, Worldwide Governance Indicators: Frequently Asked Questions, accessed October 30, 2025, https://www.worldbank.org/en/publication/worldwide-governance-indicators/frequently-asked-questions.
[31] Daniel Kaufmann and Aart C. Kraay, Worldwide Governance Indicators: Methodology and 2024 Update (World Bank Group, 2024).
[32] Ibid.
[33] Eurostat, "Real GDP per capita (SDG 08_10)," accessed October 30, 2025, https://ec.europa.eu/eurostat/databrowser/product/page/SDG_08_10.

additional proxy for country's administrative capacity and as well it measures the level of economic development of each Member State. Again, during the data manipulation with Python, the year 2025 was created by copying the data from 2024, in order to ensure consistency. At this point, all the data had been collected and merged into a single dataset containing the key explanatory variables for the upcoming analysis.

3.8 Creating the Dependent Variable and Cleaning the Data

Before performing the regressions on the data, a series of final adjustments were necessary. These included data cleaning operations, such as labeling and harmonizing true/false values into Boolean 0–1 format. The more substantial task concerned the creation of three new variables, starting with the dependent variable. As explained before, the time interval between each directive's transposition deadline and the publication date of the corresponding national measure was calculated and stored as a continuous variable. This time difference was expressed in days, weeks, months and could take both negative and positive values indicating early or late transposition, respectively. This preliminary formula inspired by Steunenberg and Rhinard (2010) was useful to have an initial idea of how compliance works across countries. However, since the empirical analysis was designed as a survival analysis, specifically using the Cox proportional hazard model, having negative values scored in these variables was not compatible with the assumption of the method. In fact, the Cox model specifically requires non-negative values, as it estimates the probability of an event occurring over time, starting from a fixed origin (in this case, the transposition deadline). Negative values stating events occurring "before time zero", are therefore not admissible under this model. Although the study cited above uses weeks to measure transposition time, this analysis uses months instead. Given the long period (2004–2025) and the dataset structure, is easier to work with a measure in terms of months and still having meaningful variation in the model. A second variable created during the dataset preparation phase was a time-varying covariate indicating whether each national transposition occurred before or after the launch of the European Green Deal on December 11th, 2019, a date that was also used to define the threshold for this dichotomous variable. Therefore, it was generated a dummy variable named

Post Green Deal that took the value 0 if transpositions published before this date and 1 for if occurred after. This variable has a key role in the analysis as it distinguishes the implementation dynamics before and after the formal adoption of the Green Deal agenda, allowing the model to assess if the pace of transpositions systematically changed following this policy shift. During the process of collecting and verifying the national transposition dates, it became clear that federal countries such as Austria, Belgium, and Germany often had a higher number of implementing acts per directive. Especially in the Austrian case where several missing transposition dates were not found in the national official journal but in the gazettes of the Länder. These observations are consistent with findings in the literature that highlight how federal institutional structures can make the implementation process more complicated as the number of actors involved increases. As example, Franchino and Borghetto (2010), studies for the role of subnational authorities in the EU law implementation. In their findings, the authors show that federalism can significantly influence transposition dynamics and performance. Therefore, a new dummy variable was created to identify federal countries. To this variable was assigned the value 1 for Germany, Austria, and Belgium, and 0 for all other member states. The choice to include only pure federal systems and not countries with significant regional autonomy such as Spain and Italy was taken since the goal was not to capture all forms of decentralisation, but only those in which subnational authorities not only implement but co-legislate and even veto transposition measures. To better study for how institutional differences affect the transposition of directives, it was added the variable *Subnational Competence*. This dummy variable identifies the countries where subnational authorities (like federal states) are responsible for implementing the EU directives. This variable has the role of distinguish between the decentralisation based on the country's structure (federal) and the other based on who is the authority that actually is responsible for the transposition of the directives. In this case, the value was set to 1 only for Austria and Belgium, as they were the only countries in the dataset where the EUR-Lex data clearly indicated, through the name of each act, that the transposing measures were directly adopted by subnational entities (i.e., the Länder in Austria and the Regions/Communities in Belgium). For all other countries, the value was set to 0.

Finally, the dataset containing all the key variables required to estimate the survival models and test if there has been a change in terms of pace in transposition since the European Green Deal has been announced. The dependent variable *Compliance Months* captures the time measured in months from the transposition deadline to the publication of national measures. To fulfill the requirements of the Cox proportional hazards model, the values occurred before the deadline were set to zero. Excluding the dependent variable, a total of 18 variables were included, covering multiple dimension that may influence the time for transposition. The first dimension captures characteristics of both the directives themselves and the structural or institutional features of the member states. It includes variables such as the variable `amending` which indicates whether a directive modifies existing legislation and *Post Green Deal*, a dummy variable equal to 1 for transpositions that occurred after the launch of the European Green Deal on December 11th, 2019. In terms of institutional features: the variable *federal*, a dummy indicating with the value of 1 if a country is a federal state. The second dimension is composed of measures that measure media attention and public opinion salience. The media salience derived from Media Cloud data is measured by *Media Salience MC* and, while *RCM Climate Salience* represents the relative climate salience index constructed from Google Trends data. The third dimension includes the government-level characteristics. These consist of several ideological indicators such as *Left-Right*, *State-Market*, *Liberty-Authority*, and *Pro-Anti EU* explained before. In addition, from the Manifesto Project were included the variables of environmental orientation: *Environmental Protection*, *Climate Energy*, *Renewables*, and *Sustainable Development*. Another structural element included, was *Technical Cabinet*, which captures whether the national cabinet in power at the time of transposition was composed of technical rather than political figures. However, it is important to note that in some cases like the Monti government in

Italy[34] or Cioloș in Romania[35], technical cabinets received strong parliamentary support. In such instances, they were treated like the political governments for the ideological classification. The fourth dimension includes six variables from the World Bank's Worldwide Governance Indicators (WGI) project, which measure governance quality and administrative capacity across Member States. These are: *Control Corruption* (use of public power for private gain), *Government Effectiveness* (quality of public services and policy implementation), *Political Stability* (risk of government instability or violence), *Rule of Law* (respect for laws and property rights), *Regulatory Quality* (ability to design and enforce sound policies) and *Voice Accountability* (citizens' ability to participate, express opinions and access free media). This dimension also includes *GDP per capita*, taken from Eurostat, which serves as a proxy for both administrative capacity and economic development. The variable *Public Concern*, created with Eurobarometer data on climate concern and filling missing values using forward substitution, accounts for citizens concern over climate change. Finally, with this final structure the dataset was now ready to conduct test and discover if the pace of transposition changed after the launch of the European Green Deal and to investigate which political, institutional or societal factors influence the timing of transposition across EU Member States.

Chapter 4: The theoretical concepts behind the model

As previously outlined, to evaluate whether a variable influences the pace of transposition either by accelerating or slowing it, it was adopted a Survival Analysis approach. The choice was driven by the fact that Survival Analysis can deal with time-to-event data (Klein and Moeschberger 2003). As previous explained, the

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[34]Gabriele Furia, "The Monti Government and the Europeanisation of Italian Politics," Sciences Po – Dossiers Bibliothèque, accessed October 12, 2025, https://dossiers-bibliotheque.sciencespo.fr/une-vie-politique-europeenne-european-political-life/monti-government-and-europeanisation-italian.

[35] "Cioloș Will Aim to Tackle Romania's Developmental Lag," Expert Briefings, December 10, 2015, Oxford Analytica, https://doi.org/10.1108/OXAN-DB207237.

dependent variable *Compliance Months*, is time-to-event variable, as it measures the number of months between the transposition deadline of and EU Directive and the date on which a Member State publishes the national measure transposing that directive. Unlike traditional regression models that focus only on whether an event occurs, the Survival Analysis incorporates the timing of the event, handles with missing observations and offers the possibility to explore which factors influence the timing of the event in question. Before going straight to the discussion of the analysis, it is important to discuss the principles of Survival Analys followed by a discussion of why this method particularly suits for the study of EU directive transposition. Finally, the final part of this chapter will introduce to the specific techniques used in this thesis, including the Cox proportional hazard model and the Kaplan-Meier curves and it will explain how these tools were employed to analyse the timing of transposition and understand what political, institutional, social factors affect and how fast countries comply.

4.1 Survival Analysis: Concepts, Functions, and Applications

A Survival Analysis is a set of statistical methods specifically designed to analyse time-to-event data, where the outcome of interest is the time until a specific event occurs. Has its roots in the biomedical field research and was developed to study patient survival time. Even if its name "survival" remains closed to its original purpose, this type of analysis has become widely adopted across different fields. For example, in engineering, the Survival Analysis (often referred to as Reliability Analysis) is used to estimate the failure time of machines and components (Lawless 2002). In economics inspects the duration of unemployment (Lancaster 1990) or the time until firms exit a market (Agarwal and Gort 1996). In sociology, it is used to examine events such as divorce (Sanizah et al. 2014). In political science, as seen in this thesis, the Survival Analysis is used to study the mining of policy implementation such as the transposition of EU Directives and cabinet survival. What makes all these cases similar to each other is their nature of time-bound process and the interest to find not only whether an event occurs but also what influence its timing. This dual focus distinguishes the Survival Analysis from other regression models that only predict the occurrence of an outcome. In practice, the Survival Analysis models the distribution of the *survival time* T, a non-negative

random variable that represents the time until an event occurs. The Survival Analysis model is composed by four core functions that characterise the distribution (Fernandes and Magalhães 2016).

First, the *Cumulative Distribution Function* $F(t)$:

$$F(t) = P(T \leq t) = \int_0^t f(s)ds$$

This function gives the probability that an event (e.g., transposition) occurs by time t. In this context, $F(t)$ expresses the probability that a Member State has transposed the directive within t months after the deadline.

Second, the *Probability Density Function* $f(t)$:

$$f(t) = \frac{d}{dt}F(t)$$

The *probability density function* describes the relative likelihood that the event occurs at exactly time t. It is the derivative of the *cumulative distribution function*. While $f(t)$ is not technically a probability (since it is a density), it is useful to understand where transpositions are more or less likely to cluster over time.

Third, the *Survival Function* $S(t)$:

$$S(t) = P(T > t) = 1 - F(t)$$

The *survival function* represents the probability that the event has not yet occurred by time t. In the context of directive transposition, $S(t)$ explains the proportion of directives that are still not transposed after t months.

Fourth, the *Hazard Function* $h(t)$:

$$h(t) = \frac{f(t)}{S(t)} = \lim_{\Delta t \to 0} \frac{P(t \leq T < t + \Delta t \mid T \geq t)}{\Delta t}$$

The *hazard function* expresses the instantaneous rate at which the event occurs at time t, given that the event has not occurred yet. For this thesis, the function quantifies the instantaneous risk that a directive will be transposed at given point in

time t, assuming it has not been transposed yet. This is central in the Survival Analysis and represents the basis of the *Cox proportional hazards model*.

4.2 The Cox Proportional Hazard Model

The Cox Proportional Hazards Model, introduced by Sir David Cox in 1972, is a statistical used in Survival Analysis to study the relationship between the survival time of individuals and one or more explanatory variables. The model estimates the hazard function, which as previously discussed, represents the probability that an event will occur at given point in time t, assuming it has not happened yet. In the medical field, survival models often assume a specific pattern for how long something takes to happen based on previous knowledge. Therefore, understanding the shape of the hazard function guides through model selection. For instance, if the risk increases over time, a model with an increasing hazard is more appropriate. While, if the risk is higher at the beginning and then drops, a model with decreasing hazard is more suitable (Lawless 1982). For the purpose of this thesis, the Cox Proportional Hazards Model was chosen because it provides a flexible framework for analysing the time until a transposition of Eu Directives occurs. The Cox Model differentiates from the fully parametric models, as it does not require to specify the baseline hazard function in advance, making it preferable in a context (like the one of directive transposition) where the hazard is unknown. The Cox model adopts a semi-parametric approach, as explained in Cox (1972), "the hazard function is taken to be a function of the explanatory variables and unknown regression coefficients". This formula leaves behind the baseline hazard $h_0(t)$ unspecified, allowing for the estimation of the effect of covariates on the hazard without assuming a particular shape for the hazard over time. Its flexibility is crucial in the context of directive transposition, where the timing of compliance is unlikely to follow a known probability distribution, and where it is important to include covariates such as political, institutional and societal variables to assess their influence over time. Kartsonaki (2021) offers a clear explanation of the Cox model.

The hazard function for an individual i at time t is given by:

$$h_i(t) = h_0(t) \cdot \exp\left(\beta_1 x_{i1} + \beta_2 x_2 + \cdots + \beta_p x_{ip}\right)$$

Where:

$h_i(t)$ is the hazard function for an individual i at time t.

$h_0(t)$ is the baseline (non-parametric) hazard function.

$\exp(\beta_1 x_{i1} + \beta_2 x_2 + \cdots + \beta_p x_{ip})$ is the exponential of a linear combination of covariates.

The key assumption of the Cox model is that hazards are proportional over time. This means that the effect of the covariates on the hazard rate is multiplicative and remains constant over time. In the context of analysing the timing of transposition of EU directives means that if one Member State has hazard of transposing a directive that is twice as high as another Member State due to political, institutional or public opinion factors, this proportional relationship remains stable over time. For instance, suppose a country governed by a cabinet with pro-environment orientation shows a 50% higher hazard of transposition (HR = 1.5). This means that at any point in time, countries with cabinet having a pro-environment orientation are 1.5 times more likely to transpose directives than countries without. While the baseline hazard (e.g., the likelihood of transposition at month 6 versus month 12) can have variations, the relative effect of the explanatory variables (e.g., salience, cabinet orientation, public opinion) remains constant over time.

4.3 The Kaplan-Meier Curves

The Kaplan-Meier estimator, introduced in Kaplan and Meier (1958), is a non-parametric method used to estimate the survival function from time-to-event data. In this model, the estimator calculates the probability that the event has not occurred by a certain point.

The Kaplan-Meier estimator is defined as:

$$\hat{S}(t) = \prod_{t_i \leq t} \left(1 - \frac{d_i}{n_i}\right)$$

Where:

d_i is the time when at least one event occurred.

t_i is the number of events (transposition) at time.

n_i is the number of directives at risk just before time.

This cumulative function accounts for the survival probability across multiple time points. Then, the result can be visualised using a Kaplan-Meier curve, a stepwise function that drops at each occurred event and remains flat between them. To explain how each drop is calculated, can be useful to look at the simplified formula of the model:

$$S_t = \frac{Directives\ not\ yet\ transposed\ at\ time\ t - Directive\ transposed\ at\ t}{Directives\ not\ yet\ transposed\ at\ time\ t}$$

The simplified function shows how that, at each time interval the curve reflects the proportion of "at-risk" directives still untransposed after the given period. The survival probability at each interval is taken by dividing the number of directive who have not experienced the event of transposition by total number of directives still "at-risk" at the start of the that interval. The directives transposed are sent out of the risk set becoming non-observable. The overall probability of survival up to a certain point in time is then obtained by multiplying the survival probabilities from each previous interval. This method is based on the idea that surviving until a given point requires surviving through all the previous intervals (Goel et al. 2010). To make a practical example with the study of the transposition of Greem Deal' directives, imagine a directive that needs to be transposed by Member States. To estimate the probability that it has not been transposed by the end of the second month, the probability of the directive of remaining untransposed after the first month is multiplied by its probability of surviving the second month, given that it was still untransposed after the first. While the probability calculated at any single interval might be not so accurate, especially when few transpositions occur at that point, the overall survival probability is more precise, providing a more stable picture of the transposition process across Member States. Kepler-Meier curves are typically represented as stepwise plots with vertical drops when an event occurs

and flat segments in between representing periods when it does not (*Figure 2*). However, in this thesis, the resulting curves appear almost smooth.

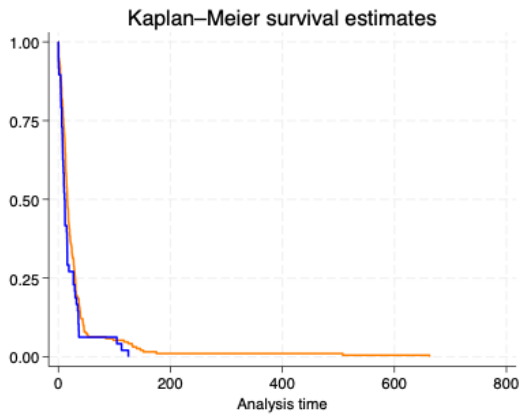

Figure 2. Example of Kaplan–Meier Survival Curve with visible stepwise pattern.

This appearance is the result of the transposition events that occur at close and regular intervals. The steps are therefore very small and frequent, giving the illusion of continuous curve rather than a staircase (*Figure 3*). Understanding this difference is essential: the smoothness of the Kaplan–Meier curve is a consequence of the distribution of data, not a change in methodology.

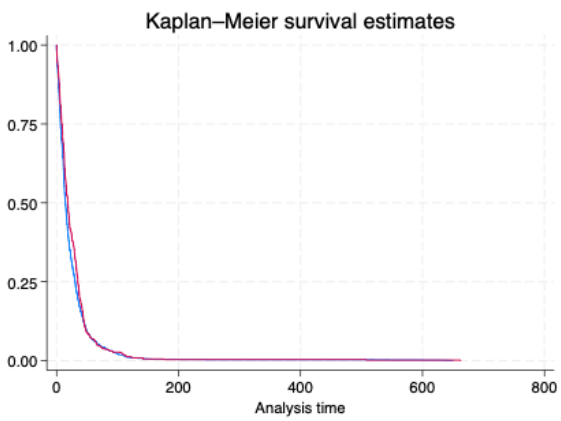

Figure 3. Kaplan–Meier Survival Curves for directive transposition (Full Sample).

4.4 Implementing Survival Models in Stata

After presented the Survival Analysis and in the specific of the Cox model and the Kaplan-Meier curves theoretically and before discussing the findings of the analysis performed, it is important to explain how the theoretical approach discussed was then translated into Stata. Before performing any regression, it was necessary not only to study the models, but also to understand clearly how they could be implemented in practice. To ensure a correct approach, this thesis relied on the official Stata manual[36], which provided detailed documentation and examples for estimating the Cox model and plotting Kaplan-Meier curves. In the dataset, this was done by checking whether the publication date was available. Based on this, a binary event indicator was created: observations with a known transposition date were coded as 1 (event occurred), while those without a date were coded as 0 (no event). Once this variable was generated, the `stset` command was used in Stata to declare the survival-time data structure. The variable *Compliance Months*, measuring the number of months between the directive's transposition deadline and the national implementation date, was specified as the time variable. The option `failure(event)` was used to define the failure event, indicating that a value of 1 in the event variable corresponds to the formal transposition of the directive. This setup is required to estimate the Cox proportional hazards model and to plot Kaplan-Meier survival curves, as it informs Stata how to treat time and events within the dataset. Once the dependent variable was declared as survival-time data using the `stcox` command, a series of Cox proportional hazards regressions were performed to explore the impact of the covariates on the timing of transposition. The analysis was with five models, each one dealing with differences between directives and countries in a different way. Some models treat these differences as fixed, meaning each directive or country is allowed to have its own specific pattern. Others treat them as random, assuming the variation follows a general pattern across cases. Using different models helps verify whether the results are consistent regardless how differences are threated. This approach allows to strength the analysis and

[36] StataCorp. Stata Survival Analysis and Epidemiological Tables Reference Manual: Release 18. College Station, TX: StataCorp LLC, 2023.

helps in revealing dynamics that might remain hidden if only a single model were used. The first model *Pooled Cox Model* is the baseline model and assumes that all directives and all countries act in the same way, without any adjustment for heterogeneity. The second model, *Fixed Effects by Directive (Stratified by CELEX)* assumes that each directive has its own baseline hazard. In simple words, the model compares the Member States within each directive, not across different directives. The third model *Random Effects by Directive (Shared Frailty by CELEX)* differentiates from the previous by using random effects instead of the fixed ones. This model assumes that differences between directives follow a common pattern. The fourth model *Fixed Effects by Country (Stratified by Country)* follows the same logic as the second model but focuses on country-level differences instead of directive-level ones, comparing directives within the same country. While the fifth *Random Effects by Country (Shared Frailty by Country)* mirrors the approach of the third model, again, applied at the country level. Instead of estimating a separate baseline for each country, it assumes that the differences between countries follow a random pattern. In addition to the Cox regressions, as said, Kaplan–Meier survival curves were used to visually display the probability of a directive not yet being transposed over time. This was done using the `sts graph` command in Stata, with the `by(variable)` option to compare groups. For boolean variables like *Amending* or *Post Green Deal*, the curves were directly stratified by their two categories (0/1). For continuous variables such as *Liberty-Authority*, a threshold (typically the mean) was used to create a dummy variable before plotting the curves.

Chapter 5: Empirical Analysis and Findings

This chapter will finally discuss the main empirical analyses conducted on the database and their findings. As explained in the last paragraph of the chapter above, the survival analysis was carried out using Stata as it offers a simpler and more intuitive environment for regressions analysis. Another advantage offered by Stata was an official and well documented manual released by the developers of the software, which specifically guides users through the correct application of

Survival Analysis, Kaplan-Meier curves and the Cox proportional hazards models. This was particularly useful in ensuring a correct translation of the theoretical model which, as examined, allows for the investigation of what influences the pace of transposition of EU directives and how. The analysis conducted on the data collected had three main goals. The first can be easily understood by looking at the title of this thesis *"Has the green deal sped up compliance? Evidence from a survival analysis of directives of the European Union"* that is finding if the launch of the European Green Deal in 2019 helped in increasing the pace at directives on environment, climate and energy are transposed by the Member States. The hypothesis is this major policy initiative, with a strong sponsorship by the European Commission might have encouraged a faster implementation. To test this hypothesis, the analysis relied on the time covariate *Post Green Deal* indicating whether each transposition occurred before or after the launch of the European Green deal[37]. The second goal was to understand which are the factors that may have influence on the pace of directives transposition and how. This analysis was conducted with the variable collected on institutional, political and social factors to see what and how influence the speed of transposition. The third analysis was to explore the difference in pace transposition across the Member States. To reach this goal, for each Member State was calculated the median transposition time to identify which countries are generally faster or slower and investigate the possible reasons behind. To address the goals, the analysis was structured around six models, each of them including variables by their thematic focus. The division in model was a decision taken to first avoid multicollinearity problems that could influence the findings and second to have a cleaner and more logical representation of the findings that would help understanding what variables influence and how the transposition of the Green Deal's directives. The first model investigates for both the first and the second goal, as it includes the institutional variables and the time covariate *Post Green Deal*, while from the second to the fifth, the models inspect for the political and societal variables. In all these models, the variables *Post Green Deal* and *amending* that assess whether a directive amends pre-existing legislation

[37] See p.13.

were used as control variables for the analyses. The sixth model investigates for the third goal by presenting the median transposition time for each Member State, identifying the fastest and the slowest countries and as well presenting what is the average for the European Union, in order to immediately see who is above or below the average. Before going into details with the findings and discussing their possible interpretations, it is useful to provide a more detailed overview of the first four models. These five models include a wide range of variables and are designed to test different theoretical expectations about what influences the pace of directive transposition. For each variable, a short explanation is given about what were the expectation about whether the factor would increase or decrease the pace of transposition. Of course, by doing the hypothesis, it was assumed that all the variables collected would influence the time depend variable and had a Hazard Ratio not equal or around the zero other than a p-value below 0.05. This step is important because it allows to assess not only the significance of the results but also whether the findings confirm or contradict the theoretical expectations developed prior to the analysis. For the sixth model, there are not hypothesis developed to prior as this models inspect for characteristics within countries and the hypotheses of a country of being slower or faster are related directly to the variables collected. To make a practical example, if the expectation for the variable "federal" were that this factor would slow down the pace, as well, the expectation from the sixth model would be of seeing federal Member States such as Austria among the slowest countries.

Model 1

The first model aims to give an answer to core question if the European Green Deal has increased the pace of the directives' transposition and to investigates the effects of institutional variables. The model includes of course the variables *Post Green Deal* and *amending* that in this case are more than simple control variables, as the model investigates for institutional effects and other institutional variables such as *Federal*, *Subnational Competence*, *Coalition Size* and *Technical Cabinet*. While *Coalition Size* and *Technical Cabinet* could be classified as political variables, their inclusion in this model is justified on institutional grounds. Both variables reflect

structural features of government formation and executive organisation that may influence administrative capacity and decision-making speed. The size of coalitions is not merely a political outcome, as it is shaped by institutional rules such as the electoral system, the degree of party fragmentation and, indirectly, by the legislative structure (e.g., unicameral vs. bicameral systems). While *Technical Cabinet*, which shows whether the executive is made of technocrats or elected politicians, can also reflect broader institutional conditions as it captures when governments operate outside normal party politics, such as during political crises where government setups may have less political authority or a weaker ability to pass laws. For each of the above-mentioned variables, were formulated the following hypothesis, based on theoretical expectations from the literature and logic of institutional functioning.

HYPOTHESIS 1.1. *If a directive is adopted after the launch of the European Green Deal, its transposition by Member States is faster.*

HYPOTHESIS 1.2. *If a directive amends existing legislation, the time required for transposition is shorter.*

HYPOTHESIS 1.3. *If a country has a federal structure, the time required for transposition increases.*

HYPOTHESIS 1.4. *If the governing coalition is larger, the time required for transposition increases.*

HYPOTHESIS 1.5. *If the government is technical rather than political, the time required for transposition decreases.*

Model 2

The second model focuses on the ideological orientation of the cabinets and uses indicators of their general political stance, such as *Left-Right*, *State-Market*, *Liberty-Authority* and *Pro-Anti EU*.

HYPOTHESIS 2.1. *If a cabinet is positioned more to the left on the political spectrum, the time required for transposition decreases.*

HYPOTHESIS 2.2. *If a cabinet favors state intervention over market mechanisms, the time required for transposition decreases.*

HYPOTHESIS 2.3. *If a cabinet prioritises liberty over authority, the time required for transposition decreases.*

HYPOTHESIS 2.4. *If a cabinet is pro-EU oriented, the time required for transposition decreases.*

Model 3

The third model examines the salience of environmental and climate issues in governments' programmes by using specific variables capturing the political support for environmental protection, climate and energy policy and sustainable development.

HYPOTHESIS 3.1. *If a cabinet expresses strong support for environmental protection, the time required for transposition decreases.*

HYPOTHESIS 3.2. *If a cabinet prioritises climate and energy policy, the time required for transposition decreases.*

HYPOTHESIS 3.3. *If a cabinet emphasises sustainable development, the time required for transposition decreases.*

Model 4

The fourth model explores the role of media and public concern regarding climate change. It includes three variables, *RCM Climate Salience*, which measures how much attention climate-related topics received from the public using Google Trends data; *Media Salience MC*, which captures how often climate topics appeared in online news using Media Cloud data and *Public Concern*, a variable from Eurobarometer surveys that reflects how concerned citizens are about climate change.

HYPOTHESIS 4.1. *If relative public attention to climate change is higher (as measured by Google Trends), the time required for transposition of EU Green Deal directives decreases.*

HYPOTHESIS 4.2. *If climate change receives more media coverage (as measured by Media Cloud), the time required for transposition of EU Green Deal directives decreases.*

HYPOTHESIS 4.3. *If public concern about climate change is higher, the time required for transposition decreases.*

The fifth model examines how the quality of governance and administrative capacity affects the transposition of Green Deal directives. It includes the six governance indicators from the World Bank's Worldwide Governance Indicators (WGI) and a structural control for economic capacity, GDP per capita.

Model 5

HYPOTHESIS 5.1: *If a country has higher Government Effectiveness, the time required for transposition decreases.*

HYPOTHESIS 5.2: *If a country has higher Regulatory Quality, the time required for transposition decreases.*

HYPOTHESIS 5.3: *If a country has higher Rule of Law, the time required for transposition decreases.*

HYPOTHESIS 5.4: *If a country has higher Control of Corruption, the time required for transposition decreases.*

HYPOTHESIS 5.5: *If a country has higher Political Stability, the time required for transposition decreases.*

HYPOTHESIS 5.6: *If a country has higher Voice and Accountability, the time required for transposition decreases.*

HYPOTHESIS 5.7: *If a country has higher GDP per capita, the time required for transposition decreases.*

Before presenting the findings, a few more methodological explanations are necessary. First, each model was estimated through an OLS regression, followed by a Variance Inflation Factor (VIF) test to detect multicollinearity. Variables with VIF values exceeding the conventional threshold (VIF > 10) were considered problematic and excluded. This diagnostic procedure helped to ensure the robustness of the Cox proportional hazard models, confirming that the explanatory variables included were not strongly correlated within each other. Second, it is important to specify that the full regression tables for the Cox Model were not included in this chapter. Instead, it was preferred to include only the outputs strictly necessary for the evaluation of the findings. Including all outputs would have used considerable space and reduced the readability of the chapter. Instead, the discussion focuses on the interpretation of hazard ratios (HRs) and the visual insights provided by Kaplan–Meier survival curves. The full output tables for all model specifications are anyway provided in Appendix A of this thesis.

5.1 Post Green Deal pace and Institutional factors

The first model of the analysis, aimed to investigate the impact of the institutional covariates on the timing of directive transposition but more important, the model aimed to answer to the main question of this thesis and finally assess if after the announcement of the European Green Deal, the pace of transposition has effectively increased.

Table 1. *Variance Inflation Factors (Model 1)*

Variable	VIF	1/VIF
Subnat. Competence	9.65	0.10
Federal	9.58	0.10
Post Green Deal	1.18	0.85
Amending	1.17	0.85
Coalition Size	1.03	0.97
Technical Cabinet	1.01	0.99
Mean VIF	**3.94**	

After the preliminary OLS regression and the VIF test for multicollinearity (*Table 1*), a high degree of multicollinearity was found between the variables *Subnational*

Competence and *Federal*. Although the VIF values were both below the conventional threshold of 10 (9.65 for *Subnational Competence* and 9.58 for *Federal*), the overlap could not be ignored and one of the two variables had to be excluded. The variable *Federal* was ultimately dropped. This decision was based on the observation that in two out of the three federal states in the sample (Austria and Belgium) Green Deal directives were transposed at the subnational level. In contrast, Germany, despite being a federal state, implemented these directives through federal-level legislation. Therefore, the variable *Federal* would only capture variation relevant to Austria and Belgium, limiting its explanatory power. In contrast, *Subnational Competence* more directly captures the for the legislative authority of transposition, regardless of formal federal status. From the analysis conducted, *Post Green Deal*, *Amending* and *Subnational Competence* variables were found being statistically significant in the results given by the five specifications of the Cox Hazard model. By contrast, the remaining variables *Coalition Size* and *Technical Cabinet* showed non-significance across all the five models. In all the specifications (pooled, fixed effects by directive and country, random effects), their p-values were constantly above the 0.05 threshold. These results suggests that factors such as the size of the government coalition or whether a cabinet is technical does not systematically influence the pace at which the Member States transpose the directives of the European Green Deal. Starting with the variable *Amending*, the results showed a strong significance across three of the five model specification. In the Pooled and country-level model, the variable showed a very strong significance with a Hazard Ratio (HR) around 1.5 and a p-value even below the 0.01 threshold. These results, strongly support for the Hypothesis 1.2 that directives amending existing legislation are transposed faster. It is important to precise that the variable *Amending* is omitted in the fixed-effects model by directive because there is no variation within each directive (all countries either transpose an amending directive or not). In the random-effects model by directive model, the coefficient is not statistically significant, this is likely due to the small number of CELEX groups and large standard errors. However, the direction of the effect remains consistent. The fact that the variable is strongly significant in the pooled model, the fixed-effects model by country and the random-

effects model by country is sufficient to support the robustness of the findings and to demonstrate that amending directives are transposed more quickly than new legislation, likely because they require fewer institutional changes and build on pre-existing legal frameworks.

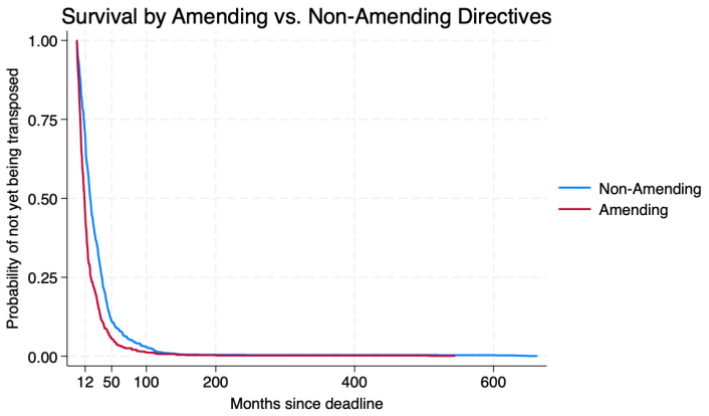

Figure 4 Kaplan-Meier curves for amending and non-amending directives

The robustness of the findings is also proved by survival curve comparing amending and non-amending directives above. *Figure 4* shows that the red line (amending) declines more, demonstrating that amending directives are transposed faster. This pattern aligns with the results from the three Cox models, where the variable *Amending* shows a HR = 1.5 and a p-value below 0.01. These results indicate that amending directives are about 50% more likely to be transposed at any given time than new directives, with. This, again, prove the idea that building on pre-existing legal frameworks facilitates a quicker compliance process. By contrast, brand new directives may require new administrative or legislative solutions and therefore more time as the legal framework within they operate could be unknown. Moving to the variable *Subnational Competence*, the results indicate a consistent negative effect on the timing of transposition across two out of three model specifications. In both the Pooled and the random-effects model by directive, the variable is statistically significant with a HR below 1 (0.88 and 0.89 respectively) and a p-value below 0.05, supporting *Hypothesis 1.3* that Subnational Competence slows down the transposition process. In the fixed-effects model by directive, the coefficient is not statistically significant, but the direction of the effect remains

negative (HR = 0.91), suggesting that the result is just random noise, but rather a stable pattern that falls just behind the threshold of significance (p = 0.091). The variable is omitted in the fixed-effects model by country, as it does not vary within countries since *Subnational Competence is* a fixed institutional feature. In the random-effects model by country, the effect remains negative (HR = 0.73) but again, not statistically significant. In this model the reason is likely due to the very small number of countries (only Austria and Belgium) for which the variable takes the value 1. However, despite its limitations, the fact that *Subnational Competence* is statistically significant in two models and displays a consistent negative direction in all others supports the robustness of this finding. It also suggests that in Austria and Belgium, when subnational authorities are directly responsible for transposing directives, the process tends to be slower by 10% to 12%, likely because of more complex coordination, administrative fragmentation, or varying regional capacities. These results will be further examined in *Model 6*, which focuses on transposition time across Member States and may provide more definitive validation of this pattern.

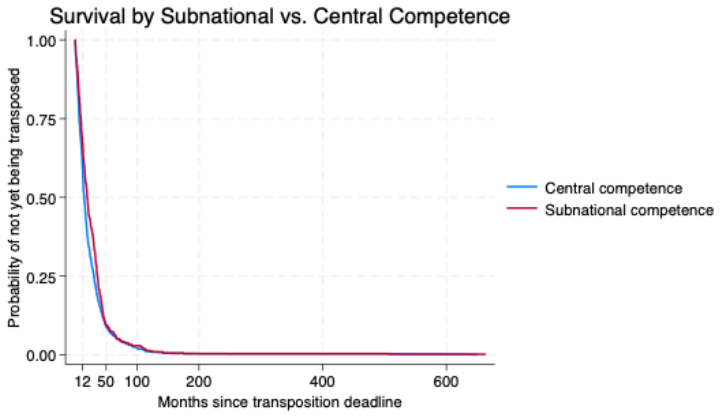

Figure 5. Kaplan-Meier Curves for Central and Subnational Competence

The survival curves show that directives delegated to subnational authorities tend to remain untransposed longer than those under central competence. This finding aligns with the regression results, where *Subnational Competence* shows a hazard ratio that despite the significance, is constantly below 1. Even if the results should be taken with caution due to the limited number of Member States (Austria and Belgium) that delegate the transposition, they likely reflect coordination problems

or administrative complexity. As mentioned before, this finding will be further explored in Model 6, which study for the transposition times across the Member States. To provide and answer to the question if the Green Deal has increased the pace of transposition, it is necessary to study in depth the variable Post Green Deal.

Table 2. *Post Green Deal effect on Transposition (Summary of Cox Models)*

	Model Specification				
	Pooled	FE (CELEX)	RE (CELEX)	FE (Country)	RE (Country)
Post Green Deal	0.987	0.032***	0.127***	1.030	1.026
	(0.0466)	(0.0083)	(0.0170)	(0.0542)	(0.0511)
N	2,334	2,334	2,334	2,334	2,334

$^{*}\, p < 0.1$; $^{**}\, p < 0.05$; $^{***}\, p < 0.01$

Table 2 isolates the values for the *Post Green Deal* variable across the five model specifications. The results show a very interesting pattern, as the variable is not significant in the pooled and country-level model, but it becomes highly significant in directive-level models (Fixed and Random effects with CELEX). Specifically, the variable gets a HR = 0.0317 in the fixed effect model stratified by directive and 0.1272 in the random effects model, in both cases the effects are statistically significant at 0.01 level. The Hazard Ratios suggest that any acceleration or deceleration in transposition associated with the Green Deal is not consistent across the Member States but rather depends on which directive is being transposed. In the specific, the fixed effect model stratified by directive accounts for built-in differences in complexity, ambition or importance between directives adopted before and after the launch of the Green Deal. The fact the variable *Post Green Deal* becomes significant only when is analysed at directive level, suggest that directives adopted after 2019 may, by their nature, be harder to transpose or face stronger political or technical obstacles. In simple words, this means that the European Green Deal may not have a uniform speed of transposition, but it might instead be affected by slower transposition for some of its more ambitious measures. To investigate further whether the Green Deal affected the pace of transposition, it was created a new model specification with the Stata command

`i.postgreendeal##i.amending`. This command treats both variables as Boolean and tells Stata to include them in the regression not only individually, but also by calculating their interaction. In other words, it tests whether the effect of being a post Green Deal directive changes depending on whether the directive is entirely new or is amending pre-existing legislation. This approach helped to better understand how the two factors of being a Green Deal directive and being an amending directive work together by checking whether their combined effect is just the sum of the two individual effects or if something different happens when both are true at the same time. This type of model with Factor Variables is used to see if the effect of one variable depends on another[38]. Therefore, by using this method, it was possible to test if the slowdown in transposition after the Green Deal depended on amending status of the directive.

Table 3. *Interaction: Post–Green Deal × Amending (Shared Frailty by CELEX)*

Variable	Hazard Ratio	(Std. Error)
Post Green Deal	0.142***	(0.023)
Amending	1.560	(1.209)
Post Green Deal × Amending	0.749	(0.195)

Notes: Cox proportional hazards with gamma shared frailty by directive (CELEX). $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The results in *Table 3*, show that the negative effect of the *Post Green Deal* variable persist even when is conducted the control for this interaction. Directives adopted after the launch of the European Green Deal are significantly slower in terms of transposition timing with a HR = 0.14 significant at 0.01 level, even after accounting for the amending status. Another interesting fact to notice is that the interaction term *Post Green Deal X Amending* is not statistically significant. This suggest that the observed slowdown affect equally both new and amending

[38] StataCorp LLC. Factor-Variable Notation: Overview. Accessed October 23, 2025. https://www.stata.com/features/overview/factor-variables/#:~:text=You%20can%20prefix%20a%20variable,the%20categories%20of%20the%20variables.

directives. Therefore, it is possible to conclude that the Green Deal has marked a clear shift in the transposition of environmental directives. To provide a practical illustration, the hazard ratio (HR) from the fixed-effects model by directive is 0.032. This means that, on average, the probability of an immediate transposition for directives adopted after the launch of the Green Deal is only 3.2% of that observed for the earlier period. In concrete terms, this represents a slowdown of approximately 30 times, equivalent to a 97% reduction in the pace of transposition. In the random-effects model, which accounts for inter-directive differences, the HR is 0.127, indicating a process roughly eight times slower (–87%). This slowdown does not probably come from the structure of the directives, but rather from the change in ambition, scope or content given by being a Green Deal's directive.

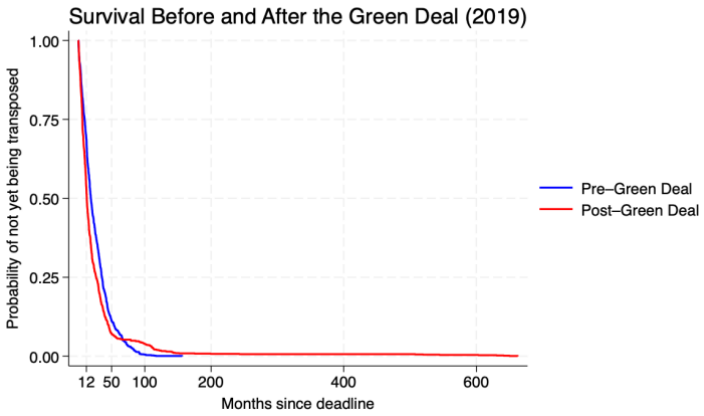

Figure 6. Kaplan-Meier curves for directives adopted after Green Deal (red) and before (blue)

The interpretation of Green Deal directives being slower is further supported by the Kaplan-Meier curves presented in *Figure 6* above. The graph comparing the curves for directive adopted before and after the launch of the Green Deal clearly illustrates a slower pace for those transposed after. The red line that represents post-Green Deal directives, declines more gradually, indicating that a great number of these directive remain untransposed for a longer period. The visual information given by the survival curves align with findings from the Cox models and reinforces the idea that the Green Deal has coincided with more demanding legislation which increase the time needed by the member states to transpose directives into national legislation acts. Another possible interpretation is that the observed slowdown may

not only reflect the increased legislative complexity or ambition but also the early signs of political resistance in some Member States. As recent events show, the Green Deal has faced increasing backlash, especially from sectors like agriculture and industry, which see its policies as too restrictive towards their economic activities[39]. This tension has translated into political debates and in some cases, open opposition to EU climate legislation. In support of this interpretation, several Member States including Latvia, Bulgaria and Spain received formal infringement procedures for failing to transpose core Green Deal directives such as Directive (EU) 2023/959 on the EU Emissions Trading System[40]. These examples suggest that beyond administrative delays, political contestation and national resistance may be contributing factors in slowing the transposition of Green Deal legislation, particularly when the perceived costs of compliance are high or politically sensitive. Moreover, since post-Green Deal directives are generally associated with slower transposition, presumably because of their complexity, the effect of accelerating the pace of transposition reinforce the idea that the complexity of an act is a key variable in terms of timing transposition. In conclusion, *Model 1* affirms that institutional characteristics play a relevant role in shaping the timing of transposition. Among the variables considered, three were found to be significant across the models: directives adopted after the launch of the Green Deal, those that amend existing legislation, and those delegated to subnational authorities. The findings suggest that the Green Deal has introduced more complex or politically sensitive directives that require additional time to be translated into national law. However, the presence of several infringement procedures against Member States for delayed or incomplete transposition of Green Deal directives may indicate early signs of political resistance in aligning national legislation with the European Green Deal framework. In contrast, amending directives are transposed more quickly,

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[39] Richard Youngs, "Europe's Green Transition Faces a Climate Backlash," Carnegie Europe, 18 September 2025, https://carnegieendowment.org/research/2025/09/climate-backlash-europe-green-transition-farmers-protests?lang=en.

[40] European Commission, "September infringement package – key decisions on energy," 26 Sept. 2024, https://energy.ec.europa.eu/news/september-infringement-package-key-decisions-energy-2024-09-26_en; European Commission, "February infringement package – key decisions on energy," 12 Feb. 2025, https://energy.ec.europa.eu/news/february-infringement-package-key-decisions-energy-2025-02-12_en.

probably because they are built on pre-existing laws and therefore need less institutional changes. The variable *Subnational Competence*, which captures whether transposition is carried out by subnational rather than central authorities, shows a consistent negative relation with transposition speed. While the effect is not always statistically significant, the direction of the effect remains stable. This suggests that delegating transposition to subnational entities may introduce delays, likely due to more complex coordination, administrative fragmentation, or varying regional capacities. Even if this finding should be interpreted with some cautio, since only Austria and Belgium fall into this category. Other features like coalition size or having a technical government did not have a significant impact.

5.2 The Role of Political Orientation in Transposition Speed

The second analysis of Cox proportional hazards models aimed to test how the ideological orientation of the cabinets affects the pace of transposition timing. The model used indicators of general political stance of cabinets along the dimensions *Left-Right*, *State-Market*, *Liberty-Authority* and *Pro-Anti EU*. The model included Post Green Deal as well and *Amending*, but these variables in this model have to role of simple control variable, so their results are not important for the topic of the secondo model. From the regressions results it appears that the clearest and most robust results is observed for the *Liberty-Authority* variable. Across all the five model specifications, the hazard ratio associated with the *Liberty-Authority* dimension remains below 1 and statistically significant. In the specific, HR stays around a value of 0.91 and the p-value is consistently below 0.05. The results show that cabinets with a more authoritarian orientation transpose directives about 9% more slowly than more libertarian ones. The observed effect suggests that cabinets characterised by authoritarian values are more committed to hierarchy, social control, status quo and may be more resistant to the implementation of European Union directives, especially those related to the Green Deal, which seems to require a significant domestic adaptation in terms of legislation. The findings support *Hypothesis 2.3*, as the coefficient for the *Liberty-Authority* variable indicates that more liberty-oriented cabinets tend to transpose EU directives more quickly. This suggests that governments prioritising individual freedoms, democratic

participation, and pluralism may be more open to communitarian policymaking and have more will to adopt new regulations proposed by the European Union.

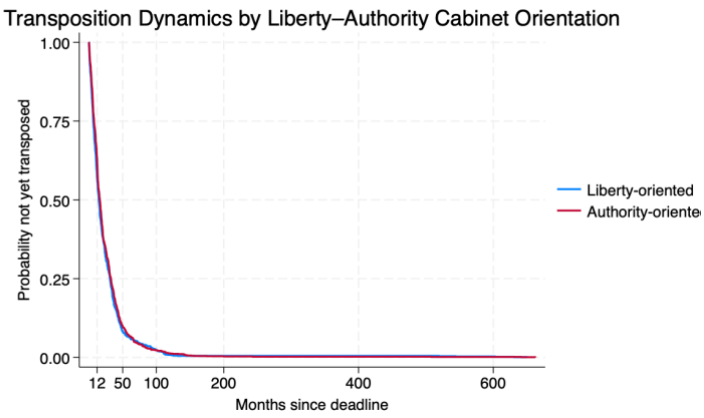

Figure 7 Kaplan-Meier curves for liberty-oriented and authority-oriented cabinets

The Kaplan–Meier survival curves, constructed by splitting countries based on whether their cabinet scored a value above or below the mean on the liberty-authority scale, offer a visual support to the finding. While the difference is slight, the curve for authority-oriented governments is consistently above the one for liberty-oriented governments, particularly in the early phases following the deadline. This suggests again that more authoritarian cabinets use more time to comply with the European Directives. Although a slight divergence between the curves, the idea that value orientation may influence compliance behavior is reinforced.

Table 4. *Effect of Left-Right Cabinet Orientation on Transposition Speed (Summary of Cox Models)*

	Model Specification				
	Pooled	FE (CELEX)	RE (CELEX)	FE (Country)	RE (Country)
Left–Right	1.058	1.082*	1.050	1.013	1.014
	(0.047)	(0.048)	(0.046)	(0.053)	(0.050)
N	2,334	2,334	2,334	2,334	2,334

Notes: Hazard ratios (Cox proportional hazards). Standard errors in parentheses.
* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

In contrast, as *Table 4* shows, the results for the variable *Left-Right* show no real significant effects on the timing transposition for Green Deal directives. Even if in the Fixed Effect model by directive, where the variable shows a HR = 1.082 and a significance at 0.05 level, suggest that right-oriented may be slightly faster to transpose the directives, this result does not hold across any other model. In fact, for all the other specifications, the Hazard Ratio is close to 1 and show no significance. This inconsistency indicates that the classical left-right, might not be a real predictor of compliance at least with the Green Deal legislation. This result breaks the traditional model of ideology orientation and tells that the fact of being left-wing or right-wing does not automatically predict a government's approach to environmental regulation, especially in the context of the European Green Deal. But this is not surprising, although environmental policies are often associated with left-leaning governments (Baranowski et all. 2024), the compliance literature has shown that a cabinet's position on the left–right dimension is not a significant explanatory factor (Toshkov et al. 2010). For the remaining two factors of this model *State-Market and Pro-Anti EU*, the results follow the pattern of the left-right orientation and do not show any significance. Across all model specifications, these variable are systematically insignificant with Hazard Ratios around 1 and relatively large p-values. This means that the attitude of governments towards state intervention in economy or towards European integration, does not affect the pace of compliance of the Green Deal directives. While the non-significance of the *State-Market* variable may be acceptable, it is indeed surprising that the test for *Hypothesis 2.4* also reveals no significant effect on transposition timing. The hypothesis that pro-EU governments would transpose directives more quickly seemed plausible, as one could reasonably expect Eurosceptic cabinets to show some degree of resistance when implementing European legislation. However, the results suggest that when it comes to implementing concrete legislation, general positions towards European integration might be less relevant than institutional, legal, or technical factors. To further explore this issue, an additional check was conducted with the infringement procedures initiated by the European Commission for two specific Green Deal directives (2023/958 and 2023/959). Infringements were launched against several countries, including Latvia, Poland, Bulgaria,

Estonia, and Spain, due to delayed or incomplete transposition. Many of these Member States were, at the time, governed by coalitions or parties generally supportive of EU integration, like the government Sánchez in Spain, Tusk in Poland and Kallas in Latvia as shown by ideological indicators from the Manifesto Project. This observation weakens the idea that opposition to the EU is a primary driver of non-compliance. Instead, it suggests that even pro-EU governments may struggle with timely transposition, possibly due to the complexity, administrative burden, or politically sensitive nature of the Green Deal directives. Overall, *Model 2* reveals that political ideology does play a role in compliance, but not in the way commonly known. The most important ideological predictor is governments' orientation over the liberty-authority dimension, rather than the conventional economic or EU-related dimensions. This finding highlights the importance of cultural values and regulatory preferences in shaping how countries respond to EU environmental rules. Rather than focusing on traditional divisions like left vs. right or pro vs. anti Eu integration, this insights suggest of looking at how governments understand state authority, individual freedom, and social control as factors that shape the pace of adoption of the ambitious climate policies.

5.3 Cabinets orientation over Environment and Climate

Model 3 examines how cabinets positions over environmental protection, climate energy policy and sustainable development affect the timing transposition of the European Green Deal directives. The four key variables of this model *Environmental Protection*, *Climate Energy*, *Renewables* and *Sustainable Development* were created from data of the Manifesto Project[41] and reflect ideological orientations of these specific issue. Like in the previous model, *Post Green Deal* and *Amending* are included as controls variables and are not interpreted here.

[41] See pp. 8-9.

Table 5. *Variance Inflation Factors (Model 3)*

Variable	VIF	1/VIF
Climate Energy	27.64	0.036175
Renewables	27.54	0.036307
Post Green Deal	1.28	0.779233
Environmental Protection	1.22	0.817349
Amending	1.10	0.910723
Mean VIF	**9.98**	

Notes: High VIF values (typically >10) indicate potential multicollinearity.

After the preliminary OLS regression, the VIF test showed in *Table 4* presented extremely high multicollinearity (VIF > 27) for the variables *Climate Energy* and *Renewables*, meaning that these two factors were strongly correlated. To address this issue, the variable *Renewables* was excluded from the final model. The decision to keep *Climate Energy was* based on its larger scope, as it captures more the policy stance toward energy transition and climate mitigation. Once *Renewables* was removed, all VIF scores went well below the multicollinearity threshold (VIF < 10), indicating no remaining collinearity concerns. Starting the discussion of the results with *Environmental Protection*, no significant effect emerged across all the model specification. From the results, the Hazard Ratio range stays around the value of 1 (e.g., HR = 1.00 in the pooled model, HR = 0.98 with directive fixed effects), and p-values remain far from statistical significance. These results suggest that the general rhetoric on environmental protection does not accelerate or delay transposition. A possible explanation is that this variable captures an often "symbolic" commitment, which does not translate into concrete or urgence acts towards the implementation of the Green Deal directives. It is the case of Sapin, where despite the Sánchez government's commitment to both European integration and ambitious climate action[42], the country received two infringement procedures

[42] Pedro Sánchez calls for unity to 'forge a State Pact to address the Climate Emergency,'" La Moncloa, October 14 2025, https://www.lamoncloa.gob.es/lang/en/presidente/news/paginas/2025/20251014-climate-emergency-convention.aspx

from the Commission for failing to transpose Directives 2023/958[43] and 2023/959[44]. As of October 2025, no official reply from the Spanish government has been submitted. This suggests that rhetorical alignment and policy ambition at cabinet level are not always quickly translated in compliance and reinforcing the idea that other institutional or technical factors may be more decisive. In contrast, the variable *Climate Energy* displays a slight tendency to slower transposition in models at directive level. For example, in the fixed effect model by directive the Hazard Ratio drops below 1 (HR = 0.93), indicating that greater climate-energy commitment suggests slower transposition around 7%. However, the effect is not robust across all the model specification, as p-values show no consistency in the pooled model (p = 0.11) and in the random effect model by country (p = 0.23), suggesting that directive-level factors may drive the effect of the variable.

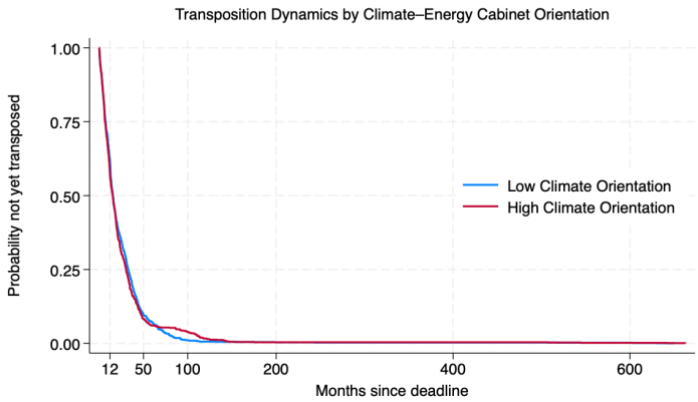

Figure 8 Kaplan-Meier curves for climate-energy cabinet orientation

The Kaplan–Meier curves reinforce the finding visually as the curves show that governments with stronger climate orientations tend to have slightly higher survival probabilities in the early months after the deadline. These results go against

[43] European Commission. Infringement Procedure INFR(2024)0050: Failure to transpose Directive (EU) 2023/958. Accessed October 23, 2025. https://ec.europa.eu/implementing-eu-law/search-infringement-decisions/

[44] European Commission. Infringement Procedure INFR(2024)0051: Failure to transpose Directive (EU) 2023/959. Accessed October 23, 2025. https://ec.europa.eu/implementing-eu-law/search-infringement-decisions/

Hypothesis 3.2, which expected that governments focused on climate and energy would transpose EU directives more quickly. A possible is that these governments may take more time to make sure the new rules fit well with their climate plans, which can make the process slower even if they support the policy. The strongest and most consistent finding of this model is given by the sustainable development orientation. Across all the five model specifications, the coefficient for the variable *Sustainable Development* are statistically significant, with Hazard Ratios raging from HR = 0.89 for the directive fixed effect model to HR = 0.92 for the pooled model. This suggests that cabinets with a stronger commitment to sustainable development are consistently slower in transposing EU directives, again in contrast with the initial *Hypothesis 3.3*, which, *like Hypothesis 3.2* regarding *Climate Energy*, expected that governments prioritising sustainable development would implement EU directives more quickly.

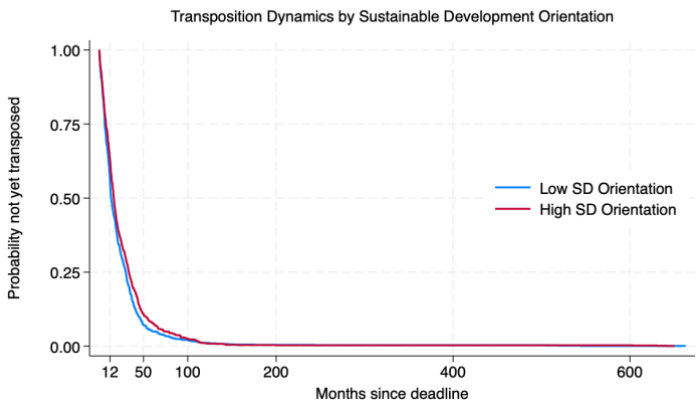

Figure 9 Kaplan-Meier curves for sustainable development cabinet orientation

The Kaplan–Meier survival curves provide further supports to the findings from the Cox models, by showing the curve for countries with high sustainable development cabinet orientation having slower transposition throughout the time window. Unlike the climate–energy variable, this effect holds across almost all the specifications, pointing to a possible structural pattern. A plausible explanation is that sustainable development implies inter-ministerial coordination, longer integration and a need to align transposition with national sustainability strategies. This complexity can therefore delay compliance, even when the alignment is high. Overall, the results

from *Model 3* challenge the assumption that governments with more progressive environmental and climate agendas are faster to comply with EU environmental legislation. This does not suggest a lack of commitment but rather, it shifts the focus to the ambitions and goals of these cabinets, which may slow down transposition due to the demanding nature of the implementation process. These findings highlight the importance of looking beyond ideological labels and considering how governance style, institutional coordination and policy complexity shape the relationship between political preferences and legal compliance.

5.4 Media Attention and Public Concern

The fourth model investigates how climate salience and public concern towards climate change affect the speed of transposition of Green Deal directives. Following the preliminary OLS regression, the media salience variable (measured using Media Cloud data) was excluded from the analysis due to limited temporal coverage (2014–2015). Including this variable would have resulted in the loss of nearly 1,000 observations. On the other side, *RCM Climate Salience*, measured through Google Trends and calculated by dividing the average annual interest for the topics "climate" and "green deal" by that for "economy"[45], showed a statistically significant positive effect in the pooled model (HR = 1.45, $p < 0.05$) suggesting an increase in transposition speed of 45% in line with *Hypothesis 4.2*. However, the effect disappears in all the other specifications for fixed and random effects. This suggests that the effect is probably driven by cross-country effects rather than real responsiveness to climate salience. In other words, relative climate salience may reflect short-term political attention rather than a lasting commitment to implementation. The third variable of the model *Public Concern*, which captures public concern about climate change based on Eurobarometer surveys[46], emerges as statistically significant predictor of slower transposition. Across all specifications, the hazard ratios for the *Public Concern* variable remain well below 1, ranging from 0.09 to 0.14 and are statistically significant at the 0.001 level. This indicates that in countries where public concern about climate change is higher, the

[45] See pp. 6-8.
[46] See pp. 10-11.

likelihood of transposing a directive at any given point in time is about 90% lower, implying a substantial delay in the implementation of Green Deal directives. These findings are in contrast with *Hypothesis 4.1* that expected higher public concern accelerates compliance due to pressure on governments. One plausible interpretation is that a higher concern from the public, brings governments to approach transposition more carefully by integrating new EU requirements with broader national climate strategies. This could involve more consultation, coordination and institutional adaptation, which slow down the transposition process.

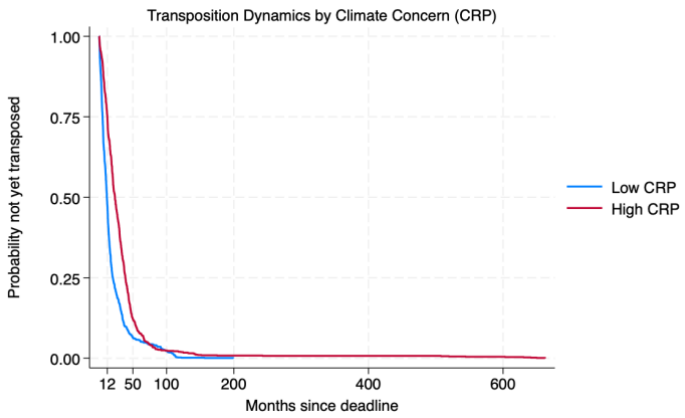

Figure 10 Kaplan-Meier curves for transposition by level of climate concern (PUBLIC CONCERN)

The Kaplan-Meier survival curves for the climate concern variable *Public Concern* show a clear and persistent gap as countries with high level of concern exhibit higher survival probabilities and take more time to transpose the directives, particularly in the early months. The visual pattern aligns with the regression results and strengthens the interpretation that higher public concern slows down the timing of transposition. Overall, these findings suggest that public attention or concern does not fasten implementation. By contrary, they suggest that when citizens are more concerned about climate change, governments may adopt a more approach to transpose EU legislation with national strategies and this can delay the process.

5.5 Public administration performance and Political stability

To study the role of administrative and political quality in the transposition of EU directives, *Model 5* includes of state stability alongside GDP per capita and standard controls. *Model 5*, aimed to include all the six Worldwide Governance Indicators, however the VIF test of the preliminary OLS regression displayed in *Table 6* below, revealed high collinearity between for four out of the six WGI indicators.

Table 6. *Variance Inflation Factors (Model 5)*

Variable	VIF	1/VIF
Rule of Law	17.51	0.057115
Government Effectiveness	14.20	0.070441
Control Corruption	13.95	0.071668
Voice and Accountability	12.12	0.082480
Regulatory Quality	7.63	0.131034
GDP per capita	3.24	0.309025
Political Stability	1.90	0.527267
Post Green Deal	1.31	0.762688
Amending	1.18	0.847953
Mean VIF	**8.12**	

Notes: High VIF values (typically >10) indicate potential multicollinearity.

In particular, *Rule of Law* (VIF = 17.51), *Government Effectiveness* (VIF = 14.20), *Control of Corruption* (VIF = 13.95) and *Voice and Accountability* (VIF = 12.12) showed the highest collinearity values. While the choice of variables was driven by the VIF scores, it also had to consider their theoretical relevance of the variables in question. In the final model, alongside the control variables, the indicators Regulatory Quality, GDP per capita, Political Stability, and Government Effectiveness were included, even though the latter displayed a relatively high collinearity value in the preliminary analysis. The variable *Government Effectiveness* was kept as it is the core explanatory factor, as it captures administrative capacity and policy implementation, *Regulatory Quality* complements it by reflecting the country's ability to design and enforce effective regulations, *GDP per capita* serves as a proxy for economic resources and bureaucratic development and *Political Stability* controls for the broader political context. *Government Effectiveness* showed a consistently positive and statistically significant effect in both the fixed effects model by country (HR = 2.03, $p < 0.001$) and the random effects model by country (HR = 1.75, $p < 0.001$), suggests that

countries with stronger administrative capacity tend to transpose Green Deal directives about twice as fast as those with weaker bureaucratic systems, thereby supporting Hypothesis 5.1. However, the fact that *Government Effectiveness* is only significant in country-level models suggests that its impact is structural and captures only differences across countries rather than across directives. This reinforces the idea that administrative effectiveness shapes countries' compliance patterns in general, rather than influencing transposition to individual directives.

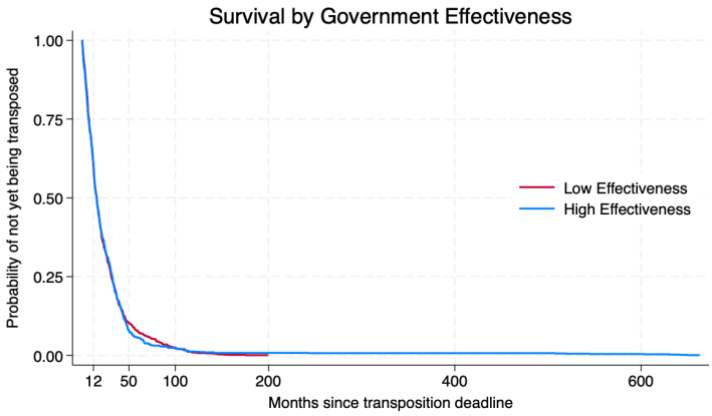

Figure 11. Kaplar-Meier Curves by Government Effectiveness

Although the difference between the two survival curves are small (probably due to the non-significance in the pooled and directive-level models), *Figure 11* further support the country-level Cox regressions results, as the blue line (high effectiveness) drops faster than the red line (low effectiveness), especially in the first 100 months after the transposition deadline, indicating that countries with higher government effectiveness transpose directives faster. Similarly, Political Stability shows a strong positive effect in both country-level models (HR = 2.91 and HR = 2.52), meaning that in more stable political contexts, the probability of transposing a directive at any given time is almost three times higher than in less stable ones.

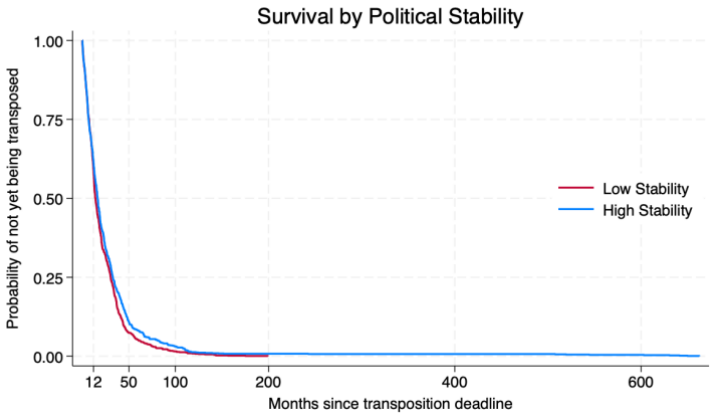

Figure 12. Kaplan–Meier Survival curves by political stability

The Kaplan–Meier curves in *Figure 12* shows little difference in transposition speed between countries with high and low political stability, but it suggests that countries with high political stability are slightly slower. However, this is in contrasts with the Cox regression results, which show the opposite. This inconsistency mostly originates from the methods: the Kaplan–Meier plot is unadjusted, while the Cox model controls for other factors like administrative capacity and economic resources. The origin of the problem, it is mostly due to the methodological differences. The Kaplan–Meier curve, in fact, provides an unadjusted, bivariate view, while the Cox models control for other factors and account for differences between countries and directives. By including fixed or random effects, it is possible to isolate the effect of *Political Stability*, correcting for the fact that some countries are slower or some directives more complex, regardless of political conditions.

Table 7. *Effect of GDP per capita on Transposition Speed (Summary of Cox Models)*

	Model Specification				
	Pooled	FE (CELEX)	RE (CELEX)	FE (Country)	RE (Country)
GDP per capita	1.0000	0.99999**	0.99999**	0.99995***	0.99997***
	(0.00002)	(0.00002)	(0.000002)	(0.000009)	(0.000005)
N	2,325	2,325	2,325	2,325	2,325

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

As shown in *Table 7*, the variable *GDP per capita* is statistically significant in most Cox model specifications but its HRs are extremely close to 1 (e.g., 0.99 in the RE Country model) suggesting a minimal impact. A possible explanation of this effect, it is probably the overlapping between *GDP per capita* and *Government Effectiveness*. In short, while the variable is significant, the effect is largely explained by bureaucratic quality and state capacity. In sum, this paragraph shows that administrative capacity and political stability are important elements in explaining the transposition timing, while the economic development (GDP per capita) plays only a minor role, likely because its effects are already captured by the other indicators. *Regulatory Quality* consistently displayed non-significant p-values, reinforcing the idea that these indicators capture overlapping dimensions of bureaucratic and institutional performance.

5.6 Beyond the Models: A Comparative Look at Member State Performance

While the previous models used a statistical approach to estimate the determinants of transposition speed, *Model 6* adopts a more qualitative and comparative perspective. This shift is both methodological and theoretical; by examining compliance at the level of individual Member States in fact, the model offers a more in-depth look for the two main schools of thought on compliance: the Management and Enforcement schools. Analysing countries' performance helps to reveal whether delays are mainly due to capacity constraints, as the management school suggest, or to political resistance and strategic behavior, as argued by the enforcement approach. It does that by looking at the median time requires to comply with the Green Deal directives across the Member States. Even if the statistical models already provide robust insights about what accelerate or decelerate transposition, comparing medians at country level, offers an intuitive and empirically valid photography of how fast each Member State tends to transpose a directive from the Green Deal. In simple words, it makes visible the timing variation across Member States and helps identify those countries driven by administrative constraints or political will, as well as those particularly efficient in transposing Green Deal directives. Ultimately, it also allows for a more comprehensive

understanding of transposition timing across Member States in the implementation of the European Green Deal. To analyse for national differences in timing transpositions, it was chosen the median over the mean. This decision was taken to as the median, unlike the mean, is less sensitive to extreme values. In the context of directive transpositions in fact, even a small amount of late transposition, especially those happened several years after the deadline, can distort the average upwards and change the overall picture of a country's performance. By contrast, using the median allows to look for the usual transposition time, reflecting the middle value in the distribution and presenting a more precise photograph of national responsiveness. This choice is particularly important in a dataset like the one used for this thesis, which cases where the directive has not been transposed yet. In this kind of situations, calculating the mean can be methodologically problematic, as it requires to assume values for incomplete cases. The median instead, for its nature of being a non-parametric measure, is less affected as it censors this missing data and allows to have data that are not distorted, even in the presence of ongoing or unreported transpositions.

Table 8. *Median Transposition Time (in Months) by Country*

Country	Median Months
France	6.6
Greece	7.5
Netherlands	9.3
Italy	9.6
Malta	10.9
Romania	11.0
Sweden	11.2
Slovakia	11.4
Lithuania	12.1
Germany	12.6
Finland	12.7
Latvia	12.8
Luxembourg	13.0
Ireland	14.3
Slovenia	14.4
Spain	14.5
Hungary	16.1
Estonia	16.2
Portugal	17.0
Bulgaria	18.0
Belgium	18.8
Cyprus	19.2
Austria	19.5
Croatia	20.4
Denmark	21.3
Poland	26.4
Czechia	30.4

Table 8 presents the median number of months required to transpose Green Deal directives across all the 27 Member States. The data reveal a clear variation within

countries as some comply very quickly, in less than 7 months and others require over 25 month with Czechia being the slowest with a median of 30 months. To facilitate the interpretation, Member States can be grouped into three categories based on their national medians. Fast transposers are those countries that transposed Green Deal directives in less than 12 months on average, suggesting strong administrative capacity, or highly centralised legal systems. Moderate performers fall within the range of 12 to 18 months and reflect more balanced institutional conditions, where effective legal frameworks are combined with some degree of political turnover or coalition complexity. Finally, slow transposers are those whose median time to transpose exceeds 18 months, pointing to potential structural problems, lower prioritisation of the Green Deal, or broader issues of administrative coordination and political alignment. The first thing that can be easily explored further is how many national acts each countries usually adopts to comply for a directive. This is displayed by *Table 9* below, which adds useful information to the analysis of timing transposition at Member States level.

Table 9. *Average National Acts per Directive by Country*

Country	Mean number of national acts per directive
Austria	34.08
Belgium	19.25
Bulgaria	8.62
Croatia	12.23
Cyprus	6.58
Czechia	12.38
Denmark	15.92
Estonia	9.31
Finland	15.54
France	13.62
Germany	5.17
Greece	3.54
Hungary	14.77
Ireland	4.46
Italy	3.15
Latvia	9.92
Lithuania	13.69
Luxembourg	4.38
Malta	4.00
Netherlands	9.38
Poland	11.00
Portugal	7.15
Romania	7.09
Slovakia	3.42
Slovenia	5.83
Spain	6.54
Sweden	9.38

From the average number of acts per directive it is possible to have an idea of how simple or complex national procedures are. Member States that are able to transpose with few acts may have a more centralised system or they can be more efficient in

translating EU laws into national law. On the other hand, countries that need many different national acts might either be doing are more detailed transposition or they may have more fragmented or weaker administrative system that can slow down the time necessary for compliance.

Table 10. *Average Government Effectiveness and Political Stability by Country*

Country	Government Effectiveness	Political Stability
Austria	1.53	1.00
Belgium	1.38	0.62
Bulgaria	−0.05	0.29
Croatia	0.58	0.63
Cyprus	1.07	0.49
Czechia	0.99	0.98
Denmark	1.96	0.94
Estonia	1.14	0.68
Finland	1.99	1.15
France	1.36	0.39
Germany	1.38	0.72
Greece	0.34	0.11
Hungary	0.56	0.74
Ireland	1.48	1.00
Italy	0.43	0.48
Latvia	0.77	0.51
Lithuania	0.90	0.77
Luxembourg	1.74	1.24
Malta	0.88	1.01
Netherlands	1.75	0.90
Poland	0.51	0.68
Portugal	1.04	0.84
Romania	−0.13	0.31
Slovakia	0.58	0.80
Slovenia	1.05	0.89
Spain	1.01	0.15
Sweden	1.74	1.01

Table 10 introduces the indicators regarding state performance used in *Model 5*: government effectiveness and political stability. These measures, taken from the Worldwide Governance Indicators (WGI), provide a comparative overview of the administrative and political quality across EU Member States. Government effectiveness refers to the quality of public services and of policy formulation and implementation[47], while Political stability, measures the perceived likelihood of pollical instability and/or politically motivated violence, including terrorism[48]. From the values displayed in the table, is possible to notice a significant variation among the Member States. Northen and Western countries tend to have better score in both the indicators, while Central and Eastern countries display lower levels of administrative quality and political stability. These value will help in understanding

[47] See p. 13.
[48] Ibid.

if the median for a country can be explained through administrative quality and political stability. The table shows considerable variation among EU countries. Northern and Western European states tend to score higher on both indicators, while several Central and Eastern European countries display lower levels of administrative quality and greater political volatility. These differences are relevant for understanding why some countries transpose EU directives more efficiently than others. In short, *Table 8* shows the median transposition time by country, revealing significant variation in how fast Member States implement Green Deal directives. *Table 9* displays the average number of national acts per directive, providing an insight into the complexity or fragmentation of national transposition processes. In the end, *Table 10* presents the average values for government effectiveness and political stability, highlighting differences in administrative capacity and institutional resilience among the Member States. After this explanation, the paragraph will introduce the most interest findings, selected by considering the country's name and the data among the three tables.

Fast unitary states Italy and France

A preliminary interpretation for countries that have a fast transposition such as Italy (9.6 months), can be given because of the systematically compliance with a limited number of national acts (on average 3.54 per directive). France, despite achieving an even faster transposition (6.6 month), it relies a much higher number of acts per directive (13.62), suggesting that its speed does not originate from the number of national acts but rather it may come from its strong administrative capacity, as reflected in *Table 10* with its strong government effectiveness score of 1.36 and likely, from the centralised administrative tradition, capable of managing complex legislative outputs efficiently.

The peculiar case of Spain

A particularly interesting case is Spain, already discussed in relation to its political and institutional characteristics. Despite having a relatively low average number of transposition acts (11 per directive), the country records a median transposition time of 26.4 months, but according to *Table 10*, Spain scores only 1.01 in government

effectiveness and 0.15 in political stability, suggesting that administrative capacity and political context may be the reason behind its delays.

Table 11. *Government Effectiveness and Political Stability in Spain (2005–2023)*

Year	Government Effectiveness	Political Stability
2005	1.50	0.19
2009	0.94	−0.47
2010	0.94	−0.32
2011	0.97	0.02
2012	1.06	−0.03
2013	1.17	0.01
2016	1.09	0.40
2017	1.00	0.27
2019	0.97	0.29
2020	0.85	0.41
2021	0.91	0.51
2022	0.92	0.32
2023	0.75	0.29
2024	0.92	0.32
2025	0.75	0.29

To better understand the administrative and political situation behind Spain's delays, *Table 11* presents the scores for *Government Effectiveness* and *Political Stability* from the Worldwide Governance Indicators of Sapin, covering the years in which transpositions were recorded. Spain's *Political Stability* score has shown a modest improvement in recent years, rising from negative levels in the early 2010s (e.g., – 0.47 in 2009) to positive but still limited values in the early 2020s (e.g., 0.51 in 2021). However, Spain's scores remain lower respect to the more stable countries. This likely reflects the historical political uncertainty in Spain, due to regional separatism, such as the Basque movement and the lasting tensions with Catalonia after the 2017 independence referendum. The country never showed high levels of political even in the recent years, in 2023, in fact, Spain scored 0.29, a slight drop from the 0.32 of 2022. This trend reinforces the interpretation that institutional and political uncertainty persisted even throughout the Sánchez era, despite the continuity of his governments. Since 2019, the governments led by the leader of the PSOE, have showed positive positions towards EU integration and climate action. Nevertheless, the country received two infringement procedures for not transposing two Green Deal directives[49]. This highlight that political alignment with EU goals, while important, it is not sufficient. Delays in timing transposition may come less from a lack of political will and more from institutional problems,

[49] See p. 39.

limited administrative coordination and persistent uncertainty in the political environment. The Spanish case reinforces the idea that stable institutions and effective governance are fundamental for ensuring compliance with Green Deal directives, even when governments are ideologically aligned.

Federal Structures and Timing in Austria, Belgium and Germany

A different pattern emerges when examining countries with a federal system that suggest a highly decentralised structure, such as Austria and Belgium. Both these countries fall into the group of slow transposers, with relatively high median transposition times. By looking at *Table 9*, it is impossible to not notice that Austria is, by far, the country with the highest number of legislative acts per directive, with an average of 34.08. While Belgium takes the second place with 19.25. The explanation behind it is simple as these two countries have in common the fact of being federal states. This pattern aligns with the results from the survival models with the *Federal* variable not being statistically significant, but the more precise *Subnational Competence* dummy, showing slower transposition with Hazard Ratios below 1 across the models. The highlighted situation perfectly reflects for the procedural complexity associated with federal system, when implementation responsibilities are in charge of the federal entities. The situation of Austria was clear even before the analyses, as during the data collection process, it became evident that many Austrian transposition measures were published not at the national level but within the official journals of the Länders. Belgium, as well, showed a similar pattern since the responsible for the implementation of the Green Deal directives are its federal regions. Moreover, both countries recorded high scores for Government Effectiveness in *Table 10*, 1.52 for Austria and 1.36 for Belgium, confirming that their slower transposition is not due to weak administrative capacity, but rather to the complexity of their federal legal systems. These systems, infact, require coordination across multiple layers of government, making it more difficult to ensure a consistent national approach. This theory is further demonstrated by Germany, where despite the fact of being a federal country, the transposition of the Green Deal directives oversees the federal government. The country shows a median transposition time of 12.6 months, placing it among the

moderate performers, with an average of 5.17 legislative acts per directive and a score of 1.64 for government effectiveness. The case of Germany ultimately confirms that the impact of federalism on transposition timing does not depend on the formal structure of the state, but it is originated from the distribution and coordination of the legislative competences. In other words, federalism itself, is not necessarily a barrier to a timely compliance, the transposition gets slower only when managed at subnational level instead at federal or government level.

Institutional Volatility in Bulgaria

Bulgaria offers a clear example of how political instability can undermine the timely transposition of Green Deal directives. With a median transposition time of 18 months and an average of 8.62 legislative acts per directive, Bulgaria qualifies as a slow transposer. This rules out administrative fragmentation as a primary cause and instead points toward political and institutional dynamics.

Table 12. *Government Effectiveness and Political Stability in Bulgaria (2006–2025)*

Year	Government Effectiveness	Political Stability
2006	−0.13	0.39
2010	−0.07	0.36
2011	−0.07	0.30
2012	−0.03	0.38
2013	−0.01	0.17
2014	−0.11	0.08
2015	−0.01	0.01
2016	0.09	0.07
2019	0.17	0.56
2020	−0.21	0.40
2021	−0.17	0.36
2022	−0.27	0.31
2023	0.05	0.34
2024	0.05	0.34
2025	0.05	0.34

To better understand this pattern, *Table 12* presents the scores of *Government Effectiveness* and *Political Stability* for Bulgaria across the years. The data show constant low levels for both the indicators. *Government Effectiveness* remained negative or close to zero for most of the period (e.g., –0.27 in 2022; 0.05 in 2023–2025), while Political Stability fluctuated at modest levels (between 0.31 and 0.39), with drops between 2013 and 2016 and slight improvement in the recent years.This weak institutional baseline is further aggravated by the recent governments crises By looking at the dataset of cabinets, build through ParlGoV and Manifesto data, to study for the political positions of governments, it is possible to notice that

Bulgaria experienced a frequent government turnover for a long period starting 2021. During this period, many governments succeeded and some of them were technical that unlike cabinets of Monti in Italy and Cioloș in Romania[50] did not receive any parliament support[51] [52] [53]. This forced them to operate without a clear political mandate, likely limiting their capacity to pass legislation and implement policy. This unstable context likely slowed the capacity of national institutions to priorities, coordinate and comply for the complex EU directives, especially those from the Green Deal. The case of Bulgaria underlines how fragile political environments can act as structural barriers to compliance, particularly when dealing with long-term policy frameworks such as the European Green Deal.

Hungary's Delays and Political Friction

In contrast to Bulgaria, Hungary represents a complete opposite scenario since the country is experiencing a high political stability since 2010, when Viktor Orban was appointed as prime minister for the first time. Through this period, his governments had stable parliamentary majority and centralised the control over the legislative process. From this situation it is plausible to expect Hungary being between the fastest transposers. Anyway, Hungary presents a median of 16.1 with an average of acts per directive of 14.4, placing it among the slower Member States. This contradiction is analytically relevant because it suggests that political stability does not automatically ensure compliance. As reported in *Table 10*, Hungary displays 0.66 for *Government Effectiveness* and 0.74 for *Political Stability*, values that are moderate but not particularly low. Therefore, neither political turnover nor administrative lacks can fully explain the delays. A more convincing explanation suggests that the key may be the qualitative nature of governance. According to the

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[50] See p. 15.
[51] Bulgarian President Announces Second Interim Cabinet as New Party Emerges," Balkan Insight, September 16, 2021, https://balkaninsight.com/2021/09/16/bulgarian-president-announces-second-interim-cabinet-as-new-party-emerges/.
[52] Bulgaria President Calls Oct. 2 Snap Election, Appoints Caretaker Govt.," Reuters, August 1, 2022, https://www.reuters.com/world/europe/bulgaria-pres-calls-oct-2-snap-election-appoints-caretaker-govt-2022-08-01/
[53] Bulgaria Gets Caretaker Government, Seventh Election in 3 Years," RFE/RL – Bulgarian Service, August 26, 2024, https://www.rferl.org/a/bulgaria-gets-caretaker-government-7th-election-in-3-years-to-be-held-october-27/33092709.html.

European Commission's 2021 Single Market Scoreboard, Hungary recorded a transposition deficit of 1.0% and a conformity deficit of 1.6%. These number places the country above the EU average for incorrectly transposed directives[54]. This hypothesis is further supported by K-Monitor's 2023 assessment of Hungary's management of EU funds. Where are highlighted serious administrative weaknesses, including delays in project implementation, lack of planning and significant difficulties in absorbing EU resource, particularly at the municipal level. These findings suggest that Hungary's administrative system, while moderately efficient according to the scores showed in *Table 10*, may lack of capacity for the timely implementation of complex EU legislation like the one of the Green Deal[55]. In addition, the political resistance to EU goals is likely to play an important role. The prime minister Viktor Orbán, has repeatedly criticised EU climate policies, describing the Green Deal as a "utopian fantasy" and warning that such measures would drive up energy prices and "destroy the European middle class"[56]. This skepticism is also reflected in the positions of Orbán's cabinets as recorded in the dataset. His governments have a 6.61 average on the liberty-authority scale, indicating a strong preference for state control. They also show low values on the Pro-Anti EU scale (e.g., 2.55 during Orbán's third term), suggesting a persistent reluctance toward EU law. In sum, Hungary illustrates a case of "stable slowness", where moderate institutional strength is not aligned by political will. The delays do not seem to come from instability, but rather from selective engagement with EU objectives. If Spain shows how limited administrative capacity and political uncertainty can delay transposition despite strong support for both EU integration and environmental policies, Hungary presents the opposite situation: a country with moderate state capacity and a mostly stable political environment, may still delay transposition when lacking political commitment.

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[54] European Commission. Single Market Scoreboard: Hungary — December 2021. Brussels: European Commission, 2021.
[55] K-Monitor, Hungary EU Funds Assessment Q3 2023 Table, 2023.
[56] Utopian fantasy': Hungary's Orban dismisses EU climate policy plans," Reuters, October 21, 2021, https://www.reuters.com/world/europe/utopian-fantasy-hungarys-orban-dismisses-eu-climate-policy-plans-2021-10-21.

Poland shows a similar pattern to Hungary. With a median of 26.4 the country is the second slowest for timing transposition of Green Deal directives behind Czechia. Poland provides another example of how political orientation and strategic contestation over EU climate priorities can slow down transposition processes. Since 2015, Poland has been governed by the conservative Law and Justice Party (PiS), first under Beata Szydło and then, from 2017 to 2023, by Mateusz Morawiecki. These governments showed signs of Euroscepticism, as confirmed by data from the ParlGov and Manifesto Project. Both Morawiecki I and II scored very high on the *Liberty–Authority* scale (8.25), showing a strong preference for state control. Their position on the *EU Anti–Pro* scale was neutral to slightly skeptical (5.48), suggesting limited support for EU integration. In addition, their score on *Environmental Protection* was low (3.96), indicating that governments under PiS were not environmentally friendly. Together with their consistent opposition to aligning EU climate ambitions with Polish economic interests, particularly given the country's reliance on coal[57], prove that the PiS-led governments were not aligned with the main goals of the European Green Deal. Moreover, according to the last European Commission's Single Market and Competitiveness Scoreboard from December 2024, Poland recorded a transposition deficit of 1.7% (compared with the EU average of 0.8%), an average delay in transposing directives of 16.9 months (versus the EU average of 11.9 months) and a conformity deficit of 1.3% (above the EU average of 0.9%)[58]. This pattern is not only political, but also institutional. In *Table 10*, Poland scored an average of 0.51 in Government Effectiveness and 0.68 in Political Stability. These are not extremely low values, but they place Poland below top performers, suggesting a moderate state capacity. Thus, Poland does not suffer from administrative dysfunctions, but neither does it benefit from a highly capable bureaucracy. Moreover, under the PiS governments,

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[57] Poland plans new coal mines despite EU's call for carbon neutrality," Reuters, September 25, 2019, https://www.reuters.com/article/business/environment/poland-plans-new-coal-mines-despite-eus-call-for-carbon-neutrality-idUSKBN1WA1EF.
[58] European Commission. Single Market and Competitiveness Scoreboard: Poland, December 2024. https://single-market-scoreboard.ec.europa.eu/countries/poland_en

Poland experienced several clashes with the European Union. A report by Stefano Fella from the UK House of Commons Library[59] provides an overview of Poland's legal and political environment during this period. It highlights that the European Commission took legal action against Poland over issues like judicial independence, the rule of law, and climate policy. These conflicts reveal deeper political and institutional tensions with the EU, which also affected Poland's access to EU funds. In short, Poland illustrates how moderate institutional capacity and relative political stability are not enough to ensure timely transposition when political commitment is lacking. Like Hungary, Poland is a case of "stable resistance", where delays are not caused by within state problems, but by a deliberate distancing from the European Green Deal. This reinforces the point that strong institutions matter, but without political will, they are not sufficient for effective compliance with EU law, especially in the Green Deal's domain.

Czechia: The Slowest Transposition Record

Czechia with a median of 30.4 months is the slowest Member State overall. Unlike Poland and Hungary, that show a strong opposition to EU integration, the lasts country's governments do not appear to be strongly eurosceptic. The governments led by Babiš and Fiala cabinet scored relatively high values on the *EU Anti–Pro* scale (8.7 and 7.0, respectively), their position on the *Liberty–Authority* axis were less extreme (ranging from 3.7 to 6.5) and their commitment to environmental remained low. This suggests that climate policy was not a top priority. Political scepticism toward the EU's climate agenda was in fact present. In 2022, the Czech government led by Fiala has described the European Green Deal as an "existential threat" to the country[60], reflecting a broader narrative that sees decarbonisation as a risk to national industry and competitiveness. Moreover, the formation of a Babiš-led coalition in November 2025, which openly opposed the Green Deal[61], allying

[59] Stefano Fella, Poland: Judicial Independence and the Rule of Law, House of Commons Library, 13 February 2024.
[60] Tom Gosling, "Czechia in 2022: Gathering Forces," Balkan Insight, January 12, 2022, https://balkaninsight.com/2022/01/12/czechia-in-2022-gathering-forces/.
[61] Enrico Pascarella, "Repubblica Ceca: il nuovo governo populista di Babiš pronto alla guerra sul Green Deal," EUNews, 4 novembre 2025, https://www.eunews.it/2025/11/04/repubblica-ceca-il-nuovo-governo-populista-di-babis-pronto-alla-guerra-sul-green-deal/.

with eurosceptic parties, thus marking a further shift from limited commitment to open resistance. Anyway, the delays are also linked to structural and procedural weaknesses within the Czech public administration. According to the OECD, the country must deal with coordination problems across ministries, administrative fragmentation and burdensome implementation procedures[62]. The Office of the Government (OG), which had previously played a key role in horizontal coordination and strategic planning, saw its role diminished under the previous government. Specifically, the dissolution of its strategic planning team and the transfer of the Government Council for Sustainable Development to the Ministry of the Environment significantly reduced the OG's ability to design and align priorities across ministries[63]. These changes likely made it harder to coordinate Green Deal reforms across ministries. Despite relatively high scores in standard governance indicators, the huge time required by Czechia to transpose, point to a disconnection between macro-level institutional metrics and micro-level administrative performance. *Table 10* shows that Czechia scores high values for both *Government Effectiveness* (0.99) and *Political Stability* (0.97), suggesting that the overall quality of governance is not a problem. Rather than administrative problems or pollical resistance, the delays in Czechia seem to come from sector-specific administrative fragmentation. The public governance structure, although generally effective, may be not equipped for cross-cutting reforms requiring high levels of coordination. In this sense, Czechia could be exemplified as a case of "capacity without coordination", a country with strong institutional frameworks on paper, but limited steering power in practice. The case highlights the importance of strategic coordination mechanisms, sometimes invisible from aggregate scores, for a timely transposition of ambitious EU directives. In conclusion, Czechia's record as the slowest Member State to transpose Green Deal directives suggests that,

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[62] OECD (2023), OECD Public Governance Reviews: Czech Republic: Towards a More Modern and Effective Public Administration,
OECD Public Governance Reviews, OECD Publishing, Paris, https://doi.org/10.1787/41fd9e5c-en.
[63] Ibid. p. 102.

despite some administrative weaknesses, the delay is mainly driven by political resistance.

In sum, *Model 6* serves as complementary to the empirical analysis not only by focusing on country-level patterns rather than the effect of individual covariates, but also by introducing a qualitative level of analysis. By comparing median transposition times, the number of legislative act per directive and governance indicators across Member States, this model offers an insight of how some Member States comply for Green Deal directives. Overall, *Model 6* enriches the analysis by providing context "behind the scenes" that cannot be fully captured by the Cox models alone but is essential to have a complete picture of how the European Green Deal is being implemented.

Chapter 6: Conclusions

6.1 Revisiting the Theoretical Framework

This thesis has developed an analysis built on the broad debate on compliance within the multilevel governance system of the European Union. As explained in *Chapter 1*, compliance does not simply coincide with the formal transposition of directives; rather, it represents a dynamic process situated at the intersection of administrative capacity, political will, and institutional coordination. The theoretical foundation of this work was built on the material presented in Professor Fabio Franchino's lectures on EU governance at the University of Milan, as well as on key academic contributions discussed during the course and additional papers identified through personal research. The literature about compliance has traditionally identified two main schools of thought: the Management School and the Enforcement School. These two approaches do not propose opposing explanations for the non-compliance of the Member States but rather offer complementary interpretations of its causes. The Management School interprets non-conformity because of administrative limits or legal ambiguity, rather than as an expression of deliberate political resistance. From this perspective, Member States do not oppose EU law due to a lack of willingness, but rather because they

lack the technical tools or resources necessary for an effective implementation. Therefore, the solution to non-conformity it is not within the nations themselves but can be achieved by strengthening capacity-building instruments, providing technical assistance and increasing administrative cooperation, rather than by imposing sanctions. This perspective aligns with studies showing that well-staffed bureaucracies and efficient coordination mechanisms can accelerate the transposition of directives (Tallberg 2002; Dimitrova et al. 2009). Similar empirical evidence emerges from the Worldwide Governance Indicators, where higher levels of *Government Effectiveness* and *Regulatory Quality* are generally associated with faster implementation. By contrast, the Enforcement School adopts a more political and strategic perspective. Based on rational choice theory, this school sees Member States as actors seeking to maximize their utility by calculating the costs and benefits of compliance. From this viewpoint, delays and infractions are not random incidents, but rather calculated choices influenced by domestic and European incentives. When the cost of compliance exceeds the expected risk from sanctions, Member States may choose to delay or even avoid transposition at all. This perspective is framed through a strategic model that sees compliance as a two-level game between the Commission, acting as the enforcement body and the Member States, acting as potential transgressors (König and Mäder 2014). It is up to the Commission decide whether to initiate or not an infringement procedure based on its chances of success and the expected costs of enforcement, while Member States try to anticipate this decision when choosing whether to comply. These two approaches are not mutually exclusive. Several studies have shown that non-compliance often results from a combination of factors, including both structural constraints and political preferences (Mbaye 2001). Weak administrative capacity can only explain part of the variation between Member States, but institutional and political elements, such as government ideology, coalition composition, or state structure (e.g., federal systems), also play a crucial role. In addition, the introduction of the qualified majority voting (QMV) in the Council of the EU has increased the likelihood that some governments may transpose directives they did not support, transforming compliance into a process of legal adaptation rather than a purely political obligation. Building on these theoretical foundations, the

empirical analysis conducted in this thesis tested the validity of these explanations by applying survival analysis models to study the timing of transposition of the European Green Deal directives and most important, if the program had accelerated the pace at compliance is achieved. The inclusion of both indicators of bureaucratic quality and political variables, particularly those capturing government orientation on environmental policy, made it possible to capture the two main dimensions of the compliance debate. The results show that both perspectives are important explanatory parts of the process, supporting the idea of a hybrid model, in which EU governance mixes problem-solving mechanisms typical of the management approach with political enforcement tools. In this sense, the European system of compliance appears to be on an equilibrium, neither purely technocratic nor entirely strategic. The Green Deal, with its ambitious goals and strict deadlines, provided a unique test for this hybrid reality, revealing that rather than accelerating compliance, it has slowed down the transposition process. This effect is likely driven by both administrative and political dynamics. On the administrative side, the wide scope and the complexity of Green Deal directives have increased the workload and coordination challenges within national bureaucracies. On the political side, the Green Deal has become one of the most politicised policy frameworks in the EU history, dividing governments and parties over issues such as energy dependence, economic costs and national sovereignty. In several Member States, especially those more dependent on coal or governed by populist or Eurosceptic parties, the opposition to the Green Deal was then translated into delays in transposition. At the same time, this thesis has identified which covariates influence the transposition timing and how. Although some indicators were specific to the Green Deal, the analytical framework was designed to be replicable across different policy areas, providing a consistent method for studying the timing and determinants of directive transposition in the European Union. To develop this theoretical and methodological framework, it was necessary to go beyond the two main schools of thought on compliance. By drawing on a broader range of studies, this thesis identified additional explanatory factors influencing the timing of EU directive transposition. As explained in *Chapter 3*, the primary goal was to analyse

the transposition of Green Deal directives[64]. However, in doing so, the key challenge was to design a model that could also be applied to different samples of directives in future research. This also made it possible to analyse another sample of directives for comparison with the Green Deal sample, giving greater robustness to the findings and testing whether certain patterns persist or change when the focus shifts from the Green Deal to other policies. The reviewed literature provided a wide range of theoretical insights. Some studies focused on directive-level characteristics, such as the time available before the transposition deadline and the distinction between brand-new directives and those amending pre-existing legislation (Mastenbroek 2003). Other works examined additional factors, including the number of actors with veto power, the degree of interministerial coordination and the level of monitoring by the European Commission (Steunenberg and Rhinard 2010; Spendzharova and Versluis 2013). Further research incorporated political and contextual dimensions, such as government orientation, the level of political conflict and the salience of the topic covered (Zhelyazkova and Torenvlied 2009). In line with these contributions, the analytical framework developed in this thesis included different variables capturing directive-level characteristics and the broader institutional and political context. Considering also the high public and political salience of the Green Deal, were included additional variables, to measure dimensions such as media attention, public opinion and government orientation toward environmental policy. This made possible to capture the full national context in which transposition occurs. The multidimensional approach presented by the model, allowed to combine structural, administrative and political explanation, offering a more comprehensive understanding of the compliance process. Moreover, the resulting framework remained flexible and replicable, offering a methodological tool that can be applied to the study of directive transposition across different policy areas in the future.

[64] See p.11.

6.2 Empirical Findings: The Post-Green Deal effect

The empirical analysis, conducted through Cox proportional hazards models and Kaplan–Meier survival curves, provides strong evidence that both administrative capacity and political dynamics jointly shape the timely transposition of Green Deal directives. The results confirmed that the two main theoretical traditions in the compliance literature (the Management School and the Enforcement School) provide complementary explanations for Member States' delays. However, the first and most significant finding of the empirical analysis concerns the *Post-Green Deal* effect, which is the core of this thesis and addresses its central question: Has the Green Deal sped up compliance? The answer is no; it has not. The results from both the Cox models and the Kaplan–Meier curve clearly indicate that directives transposed after the launch of the Green Deal (December 11, 2019) tend, on average, to have a slower transposition[65]. This effect, captured particularly in the directive-level Cox models, suggests that the introduction of the Green Deal represents a structural discontinuity in the compliance process of environmental directives, marking not an acceleration, but rather a more complex and politicised phase. This empirical result must be interpreted carefully, as it contradicts the most intuitive hypothesis that the strong initiative of the Green Deal, supported by the Commission's leadership, with high visibility and clear timelines, would lead to faster transposition. By contrast, the *Post Green Deal* effect suggests that the reform, made the transposition of directives more costly, more controversial and more exposed to domestic political dynamics. The slowdown appears to be driven mainly by two factors. First, the greater technical complexity of the directives adopted after 2019, which is due to their higher level of ambition. Second, the political resistance in some Member States that sees the Green Deal as a threat to their economic or industrial interests. Seen from this perspective, the Green Deal offers a valuable key of lecture to understand the evolution of the compliance. It shows that the Green Deal has transformed the transposition from a something technical to a strategic and political choice, shifting the equilibrium between the two greater school of thoughts that lead the literature: the Management School and

[65] For a detailed explanation see *Model 1*, pp. 40-49.

the Enforcement School. From the perspective of the Management School, the analysis confirms that administrative capacity remains an essential condition for ensuring efficient transposition. Variables such as *Government Effectiveness* and *Political Stability* show a positive association with faster transposition, indicating that strong and well-coordinated bureaucratic systems facilitate compliance[66]. The variable *Amending*, which is significant across all models, further demonstrates that directives modifying pre-existing legislation are transposed faster, as existing legislative frameworks reduce administrative costs and uncertainty. However, the significance of the *Post Green Deal* factor highlights the limits of a pure managerial perspective. The results clearly show that since 2019, political willingness and governments' strategic calculations have played an increasingly important role, consistent with the theory of the Enforcement School. In Member States whose governments have displayed authoritarian or eurosceptic tendencies, such as Poland and Hungary, the slowdown in transposition timing appears to come from a deliberate choice to resist the EU's environmental policies. In such cases, delayed transposition represents a clear sign of political will rather than a consequence of limited administrative capacity. In sum, the Green Deal has transformed compliance from a bureaucratic exercise into a political test for the European governance, a process in which administrative efficiency, institutional pressure and political will converge to determine how and when European directives are implemented.

6.3 Empirical Findings: Model by model

After having inspect for the results of the Post-Green Deal effect, it is time to do so for the results from the six models that allow to explain how political, social and institutional factors shape the pace at Green Deal directives are transposed. Always in *Model 1*[67], the variable *Subnational Competence* shows a significant negative effect, indicating that legislative decentralisation contributes to slower transposition. This is mainly due to the increased need for coordination among different levels of government. By contrast, the other two variables included in the

[66] For a detailed explanation, see Appendix A.
[67] For a detailed explanation of *Model 1*, see pp. 40-49.

model, *Coalition Size* and *Technical Cabinet*, are not statistically significant, suggesting that neither the size of the governing coalition nor the technocratic nature of the executive affects the speed of transposing Green Deal directives. From *Model 2*[68], the only significant variable that emerges is *Liberty–Authority*. Its effect on the dependent variable, *Compliance Months*, shows that more authoritarian cabinets tend to transpose directives slower. This result confirms that political culture and governmental values have a concrete impact on compliance; as governments that are more open and pluralistic are generally more cooperative with EU law. By contrast, the traditional ideological dimensions, *Left–Right* and *Pro-Anti EU,* are not statistically significant, suggesting that classic ideological polarisation matters less than governments' vision on the authority and the role of the state. The results regarding cabinets' commitment towards environmental policies in *Model 3*[69], do not support the idea that "green" governments are always faster in transposing directives. The variable *Environmental Protection* is not statistically significant, while *Climate Policy* shows a slightly negative effect. This suggests that governments emphasising sustainability and the green transition may require time to transpose directives, mostly because they put additional effort to ensure that new rules fit well with their strategies with environmental policies. Thus, even when the cabinets do support the policy's goals, the process can be slower. Once again, this finding points to the Green Deal as a complex legal and policy framework. As for *Media Salience* and *Public Concern*, the results are the opposite of the initial hypothesis. While *Media Salience* is significant in only one specification, *Public Concern* shows a consistently negative effect across all models and at a high level of significance[70]. This suggests that in countries where citizens express greater concern about climate change, governments tend to approach transposition more cautiously, taking additional time to align new EU requirements with their broader national climate strategies. An additional plausible interpretation of the results is that the slow transposition of Green Deal directives may increase citizens' perception concern towards climate risk. In other words, not

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[68] For *Model 2*, see pp.49-52.
[69] For *Model 3*, see pp.52-56.
[70] For *Model 4*, see pp.56-58.

only does public opinion influence policy, but political inertia may also generate a feedback effect on public concern. However, this hypothesis requires further testing that goes beyond the data at disposal. The fifth model[71], already discussed in relation to the main research question, aligns with the Management School of compliance. The variables *Government Effectiveness* and *Political Stability* have a significant positive effect, indicating that countries with more efficient bureaucracies and political stability transpose directives faster. *GDP per capita*, however, does not show statistical significance, likely because its effect overlaps with the other variables in the model. In *Model 6*[72], the qualitative analysis, supported by the empirical findings, confirms a hybrid model of management and enforcement, where the speed of transposition depends both on administrative efficiency (Management School) and on political will and national strategies (Enforcement School). The analysis shows that countries where transposition is managed by subnational authorities, such as Austria and Belgium and those with lower Political Stability, like Bulgaria, tend to transpose directives more slowly. Conversely, centralised countries such as Italy and France are generally faster. The cases of Poland and Hungary, whose governments are ideologically opposed to the Green Deal, further demonstrate that strategic and ideological factors can slow the transposition process, reinforcing the insights of the Enforcement School. Meanwhile, Czechia, the slowest Member State, appears to display characteristics of both theoretical traditions. Overall, these results present a coherent scenario in which compliance with the European Green Deal is neither purely technical nor purely political but rather originates from the interaction between administrative capacity and political will. The Green Deal did not accelerate the pace of transposition; instead, it transformed compliance into a political test of European governance, where speed reflects not only state capacity or efficiency but also the degree of alignment between domestic interests and conformity with EU objectives.

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[71] For *Model 5*, see pp. 58-61.
[72] For *Model 6*, see pp. 61-74.

6.4 Cross-Policy Comparsion: Green Deal and Labour Directives

To test the validity and adaptability of the model across different policy contexts, an additional empirical analysis was conducted using the same strategy on a sample of European directives regarding labour and social protection. This section aims to assess the robustness of the model developed for the Green Deal directives by applying it to a different policy are, for the purpose of comparing transposition dynamics and strengthen the empirical validity of the findings. The new dataset was constructed by filtering EUR-Lex and selecting directives under the topics employment, social policy and working conditions. Thanks to these filters, sixteen directives adopted between 2010 and 2022 were identified, covering issues such as parental leave, equal treatment, free movement of workers, transnational posting, transparent working conditions, whistleblower protection, and adequate minimum wages. These include, among others: Directive 2010/18/EU on parental leave, Directive 2014/54/EU on the free movement of workers, Directive (EU) 2019/1158 on work–life balance, Directive (EU) 2021/1883 on the European Blue Card and Directive (EU) 2022/2041 on adequate minimum wages[73]. Since the goal was to validate the theoretical and empirical model presented in *Chapter 3* and given time constraints, it was not possible to replicate the full set of variables used for the analysis of Green Deal directives. Instead, the analysis focused on a subset of structural and political variables, selected for their relevance to the two main schools of thought on compliance. These included *Amending*, *Subnational Competence* (to assess the role of decentralisation in labour directives), *Coalition Size* and *Technical Cabinet* (to test the nature of governments), the general political orientation variables *Left–Right*, *State–Market*, *Liberty–Authority*, *Pro-Anti EU* and finally, governance and administrative capacity indicators from the World Governance Indicators (WGI) and *GDP per capita* from Eurostat. Furthermore, for the variable *Subnational Competence,* it was verified whether, even in the case of labour directives, the responsibility for transposition was in charge of subnational authorities in federal states. In this case, a difference emerged compared to the Green Deal directives where the distinction between the federal and subnational

[73] The full list of directives is provided in Appendix D of this thesis.

levels was more systematic for environmental policies. In the case of labour directives, in fact, this difference appeared more limited and not homogeneous. Germany appeared to transpose the labour directives at the federal level and therefore no changes were detected. By contrast, Austria and Belgium showed a different pattern, as only some of the directives in the sample were transposed at the subnational level. These include directives on parental leave (2010/18/EU), equal treatment (2010/41/EU), free movement of workers (2014/54/EU), and the protection of whistleblowers (2019/1937/EU). Since these directives were approved by regional parliaments, the variable *Subnational Competence* was coded as 1 in the dataset, while it remained 0 for all other directives. This represents the first key difference between the two samples analysed and illustrates how the coding method allows for comparison across policy areas, capturing institutional variation within federal systems. As with the Green Deal directives, the full output tables for all model specifications are provided in Appendix B of this thesis. The Cox proportional hazards models estimated on the labour directives sample confirm the validity of the analytical framework while revealing distinct dynamics compared to the Green Deal. Although the set of variables included was more important evidence on how policy salience and legal complexity shape compliance timing. The variable *Amending* shows an opposite trend compared to the Green Deal directives. Despite its significance is limited to the models estimated at the country level, it presents a HR below 1 (≈ 0.84) with $p < 0.05$. A possible interpretation of why transposition is 16% slower for amending directives is that it often requires reopening negotiations with social partners, trade unions, and employers' organisations. This process likely increases procedural complexity and rises the time required for transposition. While the amending factor accelerate transposition in technocratic areas such as environmental policy, where a regulatory logic prevails, it appears to slow it down in sectors where negotiations are mainly domestic and involve horizontal coordination among national actors rather than supranational dynamics.

Table 13. *Effect of Subnational Competence on Transposition Speed (Labour directives)*

	Model Specification				
	Pooled	FE (CELEX)	RE (CELEX)	FE (Country)	RE (Country)
Subnational Competence	0.817***	0.791***	0.736***	1.494**	1.313*
	(0.062)	(0.070)	(0.062)	(0.283)	(0.211)
N	1,023	1,023	1,023	1,023	1,023

Notes: Hazard ratios (Cox proportional hazards). Standard errors in parentheses.
* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Interesting is also the pattern presented by *Subnational Competence* in *Table 13*, as it reflects the earlier explanation regarding the construction of the variable. Unlike in the case of the Green Deal directives, its coding was not uniform across all labour directives, since some of them were approved at the federal level in Austria and Belgium. The HRs reflect this situation, showing values above 1 in the country-level model, although the significance remains marginal, approaching $p < 0.1$. In contrast to the Green Deal analysis, where the variables *Technical Cabinet* and *Coalition Size* showed no statistical significance, the model for labour directives presents a different picture. *Coalition Size* displays a slightly negative but strong significant effect in both the pooled and directive-level models (HR ≈ 0.93, $p < 0.01$), indicating that larger coalitions are associated with slower transposition. This suggests that when politicisation at the supranational level is lower, internal government dynamics and party negotiations return to be influential. Unlike in the Green Deal model, *Technical Cabinet* shows a positive and significant effect (HR ≈ 1.80), even if this holds only in the fixed-effects models ($p < 0.05$), while in other models the effect is weaker ($p < 0.1$). These coefficients call for cautious interpretation, yet it remains plausible that for labour directives, the neutrality of technical cabinets may facilitate the transposition of such directives. This mechanism was less relevant in the Green Deal context, where politicisation was concentrated at the supranational level and shaped by broader national interests. The model examining the effect of cabinets' political orientation on the transposition speed of labour directives (equivalent to *Model 2* for the Green Deal) shows a comeback in shaping compliance of the traditional ideological dimensions.

While in the case of Green Deal directives variables such as *Left–Right* and *Pro-Anti EU* were not significant[74], in the context of labour directives both political orientation and attitudes toward European integration regain statistical relevance. This suggests that, in the absence of high technical complexity and strong supranational politicisation, transposition returns to reflect domestic political preferences. Specifically, the Left–Right variable shows a positive and highly significant effect (HR ≈ 1.20 to 1.60; $p < 0.01$), indicating that right-wing governments tend to transpose labour directives 20 to 60% faster. This is not surprising, as a similar pattern emerged with the *Climate Energy* variable in the Green Deal model, which slowed transposition under governments in favor of climate energy policies[75]. Although one might expect left-wing executives to act faster on labour-related issues, the slower pace does not an error in the model; rather, it may suggest that these governments tend to prioritise broader social dialogue and consultation with trade unions and employers' organisations before transposing EU directives. This approach, aligned with their ideological orientation, can have as consequence the slowdown of the transposition timing. The *Pro–Anti EU* variable, even if significant only in the country-level models (HR ≈ 0.80 to 0.70; $p < 0.01$), shows the expected pattern: Eurosceptic governments transpose directives 20 to 30 % more slowly, indicating that lower commitment to the EU tends to delay compliance with European legislation. This pattern reflects the logic of the Enforcement School, which highlights the behavior of governments that oppose EU integration and even if formally obliged to comply, they intentionally slow down transposition as a form of strategic delay. In the Green Deal model, however, the opposition towards EU law is not statistically significant. This was mainly due to the technical complexity of Green Deal directives, which are difficult to transpose even in countries, as in the case of Spain, with governments that expressed support to the reform and are in favour of more European integration. In this context, the results still suggest the presence of political opposition, but the lack of statistical significance of the *Pro–Anti EU* variable is likely due to the strong

[74] For a detailed explanation, see Appendix A.
[75] For a detailed explanation, see Appendix A and pp. 54-55.

technical and administrative complexity of the Green Deal directives. Therefore, these challenges may overshadow ideological factors such as pro-EU orientation. The variable *State–Market*, which is significant across all models (HR ≈ 0.80 to 0.60; $p < 0.01$), suggests that governments with a stronger state-oriented vision tend to transpose labour directives more promptly adapting the national legal framework to EU law in areas where public regulation is strong. By contrast, market-oriented governments appear to be 20 to 40% slower, likely because they prefer not to impose additional regulatory charges on companies. This effect is followed by the one of *Liberty–Authority*, which shows a similar pattern with strong significance across all five models (HR ≈ 0.80; $p < 0.01$), confirming that more authoritarian cabinets are roughly 20% slower. Compared to the Green Deal model, where *State–Market* was not significant and *Liberty–Authority* was the only ideological dimension influencing transposition[76], the results from labour directives provide further evidence of the renewed importance of political variables. This suggests that in contexts with lower supranational pressure and technical complexity, transposition once again reflects the ideological preferences of national governments. Before estimating the model for labour directives, a VIF test for multicollinearity was conducted. The test results indicated strong collinearity among the WGI indicators (*Control of Corruption*, *Rule of Law*, *Regulatory Quality*, *Government Effectiveness*, *Voice and Accountability* and *Political Stability*). To avoid multicollinearity problems while maintaining comparability with *Model 5* from the Green Deal analysis, the estimations were carried out using the same variables: *Government Effectiveness*, *Political Stability* and *GDP per capita*. Unfortunately, multicollinearity with the variable *Regulatory Quality* was too high and this variable also overshadowed the effect of *Political Stability*. Therefore, it was preferred not to include it in the model for labour directives.

[76] See Appendix A for a detailed explanation.

Table 14. *Effect Administrative Capacity and Governance Quality (Labour directives)*

	Model Specification				
	Pooled	FE (CELEX)	RE (CELEX)	FE (Country)	RE (Country)
Government Effectiveness	1.082	1.121	1.138	0.914	1.108
	(0.138)	(0.152)	(0.150)	(0.337)	(0.205)
Political Stability	0.687**	0.740*	0.717*	0.822	0.982
	(0.109)	(0.126)	(0.124)	(0.265)	(0.237)
GDP per capita	1.000	1.000	1.000	0.99993***	0.99999
	(0.00000)	(0.00000)	(0.00000)	(0.00001)	(0.00000)
N	895	895	895	895	895

Notes: Hazard ratios (Cox proportional hazards). Standard errors in parentheses.
* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

The results, from *Table 14*, reveal a completely different scenario compared to the Green Deal directives. The variable *Government Effectiveness* does not show any significance across the five model specifications, suggesting that bureaucratic quality is not a determining factor in the transposition timing of labour directives. This reinforces the idea that such measures involve lower technical complexity, unlike the Green Deal, where administrative capacity plays a central role of a fast compliance. *Political Stability*, by contrast, shows a negative and moderate significant effect (HR ≈ 0.82–0.68; $p < 0.05$), indicating that higher political stability is associated with longer transposition times. Although this result may appear counterintuitive, it could reflect a lower sense of urgency in politically stable contexts. Conversely, governments operating in less stable environments may transpose directives faster to demonstrate efficiency and avoid tensions with the European Commission. Finally, *GDP per capita* results statistically irrelevant (HR ≈ 1), which is not surprising since it serves as a proxy for state capacity and it was also non-significant in the Green Deal model, likely due to its overlap with the WGI indicators. Overall, these results suggest that while administrative capacity and governance quality were key determinants of transposition speed in the Green Deal context, they completely lose relevance in the labour sector. This confirms that the nature of the policy analysed shapes the determinants of compliance: the more a directive is technocratic and supranationally driven, the more administrative capacity matters; the more it involves social negotiation, the more domestic

political dynamics prevail. Overall, these results suggest that while administrative capacity and governance quality were crucial in determining the timing transposition for the Green Deal, within the labour sector they completely lose their explanatory role in favour of domestic political factors. This difference reflects the nature of the policies analysed: for the Green Deal, the political dynamics are largely supranational and centred on EU-level negotiations, which overshadow domestic political variables; in contrast, labour directives are less technocratic and more socially negotiable, bringing back domestic political preferences and internal government dynamics as explanatory factors.

Table 15. *Median Transposition Time and Mean of National Acts (Labour directives)*

Country	Median Months	Mean Acts per Directive
Malta	2.7	2.09
Latvia	4.8	2.40
Cyprus	4.9	2.45
Estonia	5.5	6.08
Poland	5.5	7.38
Bulgaria	5.5	6.00
France	6.5	7.92
Ireland	7.4	2.58
Hungary	8.2	9.50
Greece	9.5	2.50
Denmark	9.9	3.31
Portugal	10.0	2.75
Lithuania	11.5	10.33
Romania	12.2	5.83
Finland	12.3	10.08
Luxembourg	12.7	3.50
Spain	13.5	3.64
Netherlands	14.2	2.92
Austria	14.5	26.23
Italy	15.1	2.69
Belgium	15.4	13.73
Sweden	16.0	6.91
Croatia	16.6	9.15
Slovenia	17.8	3.42
Germany	29.3	7.90
Czech Republic	32.7	12.23
Slovakia	35.9	3.08

Table 15 shows the median transposition times for labour directives and the corresponding number of national legislative acts adopted by each country to transpose such directives. Labour directives show lower median transposition times (around 10 to 12 months) compared to the Green Deal and cross-country variation also appears to be smaller. Only three Member States display significantly longer medians: Germany (29.3 months), Czechia (32.7) and Slovakia (35.9), comparable

to the peaks found for the Green Deal in *Table 8*[77]. However, most of the Member States complete the transposition process within one year. In parallel, the number of national acts per directive decreases to an average of 2–8, with one notable peak in Austria (26.2), where, as previously shown, the federal structure continues to shape the transposition process. For the Green Deal, by contrast, median times were higher (15–16 months) and the number of acts per directive was also larger, with peaks in Belgium (19) and Austria (34), confirming the greater complexity of transposing Green Deal legislation. These structural differences between the two samples of directives suggest that politicisation and administrative structure interact differently in determining compliance speed. In the Green Deal context, politicisation occurs mainly at the supranational level: public debate, binding climate targets and tensions among Member States make transposition a political test of European governance, where each government must balance domestic pressures with EU commitments. In the case of labour directives, however, politicisation seems to be primarily domestic, reflecting negotiations between parties, trade unions and interest groups. Here, compliance follows a more procedural logic, inside a well-established legal framework but still sensitive to national political debates. In sum, while the Green Deal represents a field of competition over the direction of European integration, labour directives remain rooted in domestic negotiation mechanisms, where the conflict does not concern the European Union per se but rather the distribution of benefits and responsibilities among national actors. The empirical extension provided by the comparison between Green Deal and labour directives confirms the robustness and transferability of the model, showing that the theoretical structure based on the interaction between administrative capacity, political and institutional factors hold across different policy areas.

6.5 Future Research and Model Extensions

Future research should aim to improve and expand the model developed in this thesis, both under a theoretical and a practical perspective, to better understand the

[77] For a detailed explanation of *Table 8*, see pp. 62-63.

mechanism that rule the pace and the quality at European directives are transposed. The analysis conducted so far, has shown that compliance is not a merely administrative or political process, but rather it suggests the existence of a more complex interaction between bureaucratic capacity, institutional coordination and strategic incentives. The future direction of the research should focus on refining and expanding the model developed in this thesis, both theoretically and empirically, to achieve a deeper understanding of the mechanisms that determine the speed and quality of the transposition of European directives. The analysis has shown that compliance cannot be seen as merely administrative or political proces, but rather as the result of a complex interaction between bureaucratic capacity, institutional coordination and strategic incentives. For this reason, the model presented in this thesis should not be considered as a final outcome, but rather a starting point that can be further upgraded. This would contribute to build a broader comparative framework for understanding how directive transposition operates across different policy areas. At the same time, extending the model to study of the transposition of the European Green Deal could help identify new structural, political and sectoral factors that shape compliance behaviour for that topic. In this sense, this paragraph outlines possible directions for future research and new factors that could enrich the model, deepening the study of compliance for the Green Deal but also for other areas of EU policy. A first possible extension would be to study the complexity of directives. As already highlighted by Mastenbroek (2003) and Steunenberg and Rhinard (2010), legislative density and procedural complexity are key factors in influencing compliance time. More elaborate directives, containing numerous articles and multiple references, require greater efforts of interpretation and institutional coordination. Future research should therefore be able to quantify this complexity through objective indicators, such as the number of articles or the overall length of the text. This would allow for a more precise assessment of how legal and administrative complexity affects transposition speed, an element that has already proved to be one of the main explanatory factors behind timing transposition in the case of the Green Deal. A second possible field of extension concerns the geopolitical and sectoral dimension. By including variables such as energy dependency on Russia and the composition of the national energy mix

(nuclear, fossil fuels and renewable sources) would make it possible to link transposition capacity to the structural energy conditions of individual Member States. With these variables, it would be possible to study how energy vulnerability and strategic priorities shape the timing of compliance for directives related to the Green Deal and the ReArm EU initiative. In parallel, including an analysis of the European legislative process into the model could help to better understand cases where delays have political or conflict-related roots. First, including data on the final vote in the European Parliament would allow to measure the degree of consensus or polarisation of each directive, providing a direct prove that political conflict is already present at the adoption stage. Second, it would be useful to add a proxy for institutional conflict between the Parliament and the Council, for instance by identifying whether a directive was approved with the second reading or after long negotiations, that could translate into more difficulties during national transposition. In the end, the main limitation of this research also represents its most important aim for future development: the limited availability of directives directly linked to the Green Deal. Since most of the Green Deal measures are still in the implementation phase, replicating this study in the future with a broader and updated sample will be an essential extension. Only with a longer time arc will be possible to verify whether the trends identified in this thesis, particularly the *Post Green Deal* effect and the role of politicisation, will stabilise or will evolve as the European Union's green transition progresses.

Final reflections

The research framework developed in this thesis has shown that compliance in the context of the European Green Deal can no longer be considered a merely administrative exercise, but rather as a genuinely political process. Through the empirical analysis based on Cox proportional hazard models and Kaplan–Meier survival curves, it has been demonstrated that the speed of transposition of Green Deal directives depends on a combination of institutional, political and administrative factors, as well as on the salience of the issue and the level of public

concern about climate change. Bureaucratic capacity remains a fundamental element, yet political will and government structure inevitably affect both the timing and the modalities of compliance. From a theoretical perspective, the results confirm the validity of a hybrid approach between the Management School and the Enforcement School. On one hand, administrative quality, measured through indicators such as *Government Effectiveness* and *Political Stability*, accelerates transposition timing and confirms management-oriented theories. On the other hand, political and ideological variables reveal that governments act with a strategic behaviour, weighing the costs and benefits of compliance, as predicted by rational choice theory. The interaction between these two logics creates a model of managed enforcement, in which the European Commission plays the role of guarantor. The Commission has shown no hesitation in intervening in cases of non-compliance, yet the speed of transposition remains heavily conditioned by the internal dynamics of single Member States. Within this framework, the Green Deal represents a major challenge for the overall governance of the European Union. The politicisation of climate policies, the technical complexity of the directives and the pressure of time-bound climate neutrality goals have all highlighted structural differences in administrative systems and the varying capacities of governments to translate EU law into national legal frameworks. Nonetheless, the multilevel governance system has demonstrated significant resilience: despite delays and political tensions, most of the Member States have continued to participate actively in the process of compliance, indicating an overall stability in the EU's governance model. Ultimately, this research invites on a broader reflection on the future of European integration. The challenge is not just how to ensure the timely transposition of directives, but also how to transform the speed of compliance into policy effectiveness. The European Union's ability to maintain coherence between its ambitions and concrete results will determine not only of the success of the Green Deal, but also of the political legitimacy of the integration process itself. In this sense, the European Green Deal has turned compliance from a bureaucratic exercise into a political test of the EU's capacity to deliver on its most ambitious goals.

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Appendix A. Regression outputs for Green Deal directives

Model 1A. *Variance Inflation Factor (VIF Test)*

Variable	VIF	1/VIF
Post Green Deal	1.18	0.8507
Amending	1.17	0.8536
Subnat. Competence	1.03	0.9737
Coalition Size	1.02	0.9759
Technical Cabinet	1.01	0.9904
Mean VIF	**1.08**	

Model 1A. *Effect of Post–Green Deal and Institutional Factors (Green Deal Directives)*

	Pooled	FE (CELEX)	RE (CELEX)	FE (Country)	RE (Country)
Post Green Deal	0.987	0.032***	0.127***	1.030	1.026
	(0.046)	(0.008)	(0.017)	(0.054)	(0.051)
Amending	1.592***	(omitted)	1.188	1.600***	1.566***
	(0.077)		(0.869)	(0.084)	(0.078)
Subnat. Competence	0.883**	0.915*	0.893**	(omitted)	0.736
	(0.044)	(0.048)	(0.046)		(0.150)
Coalition Size	1.009	1.007	1.006	1.011	1.030
	(0.014)	(0.015)	(0.015)	(0.023)	(0.021)
Technical Cabinet	1.006	0.940	0.971	0.900	0.873
	(0.155)	(0.146)	(0.150)	(0.156)	(0.144)
N	2,334	2,334	2,334	2,334	2,334

Notes: Hazard ratios (Cox proportional hazards). Standard errors in parentheses.
$^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Model 1A. *Interaction: Post–Green Deal × Amending (Shared Frailty by CELEX)*

Variable	Hazard Ratio	(Std. Error)
Post Green Deal	0.142***	(0.023)
Amending	1.560	(1.209)
Post–Green Deal × Amending	0.749	(0.195)
Subnat. Competence	0.892**	(0.046)
Coalition Size	1.006	(0.015)
Technical Cabinet	0.972	(0.151)

Notes: Cox proportional hazards with gamma shared frailty by directive (CELEX).
$^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Model 2A. *Variance Inflation Factor (VIF Test)*

Variable	**VIF**	**1/VIF**
Left–Right	6.67	0.1499
Liberty–Authority	3.71	0.2693
State–Market	3.27	0.3058
Pro–Anti EU	1.60	0.6244
Post–Green Deal	1.19	0.8419
Amending	1.17	0.8549
Mean VIF	**2.94**	

Model 2A. *Effect of Political Orientation (Green Deal Directives)*

	Pooled	**FE (CELEX)**	**RE (CELEX)**	**FE (Country)**	**RE (Country)**
Post–Green Deal	1.006	0.033***	0.135***	1.079	1.074
	(0.046)	(0.009)	(0.018)	(0.055)	(0.052)
Amending	1.606***	(omitted)	1.288	1.591***	1.566***
	(0.079)		(0.879)	(0.086)	(0.079)
Left–Right	1.058	1.082*	1.050	1.013	1.014
	(0.047)	(0.048)	(0.046)	(0.053)	(0.050)
State–Market	1.011	0.975	0.999	1.045	1.053
	(0.045)	(0.050)	(0.049)	(0.052)	(0.050)
Liberty–Authority	0.910***	0.902***	0.922**	0.928*	0.922**
	(0.024)	(0.025)	(0.025)	(0.038)	(0.031)
Pro–Anti EU	0.979	1.015	1.005	0.985	0.982
	(0.037)	(0.040)	(0.039)	(0.045)	(0.041)
N	2,154	2,154	2,154	2,154	2,154

Notes: Hazard ratios (Cox proportional hazards). Standard errors in parentheses.
$^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Model 3A. *Variance Inflation Factor (VIF Test)*

Variable	**VIF**	**1/VIF**
Post-Green Deal	1.47	0.6797
Climate Energy	1.42	0.7066
Amending	1.17	0.8562
Environment Protection	1.14	0.8809
Sustainable Development	1.01	0.9916
Mean VIF	**1.24**	

Model 3A. *Effect of Orientation towards Environmental Policies (Green Deal Directives)*

	Pooled	**FE (CELEX)**	**RE (CELEX)**	**FE (Country)**	**RE (Country)**
Post-Green Deal	0.959	0.029***	0.128***	1.088	1.056
	(0.053)	(0.008)	(0.017)	(0.069)	(0.063)
Amending	1.631***	(omitted)	1.204	1.666***	1.616***
	(0.081)		(0.889)	(0.091)	(0.083)
Environment Protection	1.008	0.996	1.000	1.014	1.010
	(0.007)	(0.007)	(0.007)	(0.011)	(0.010)
Climate Energy	0.998	0.997	0.999	0.950**	0.968**
	(0.011)	(0.011)	(0.011)	(0.015)	(0.014)
Sustainable Development	0.962***	0.974**	0.975**	0.889***	0.917***
	(0.010)	(0.010)	(0.010)	(0.016)	(0.014)
N	2,235	2,235	2,235	2,235	2,235

Notes: Hazard ratios (Cox proportional hazards). Standard errors in parentheses.
$^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Model 4A. *Variance Inflation Factor (VIF Test)*

Variable	**VIF**	**1/VIF**
Post Green Deal	1.35	0.7380
RCM Climate Salience	1.26	0.7954
Amending	1.16	0.8591
Public Concern (CRP)	1.08	0.9259
Mean VIF	**1.21**	

Model 4A. *Effect of Media Salience and Public Concern (Green Deal Directives)*

	Pooled	**FE (CELEX)**	**RE (CELEX)**	**FE (Country)**	**RE (Country)**
Post Green Deal	1.151***	0.018***	0.061***	1.190***	1.203***
	(0.059)	(0.004)	(0.009)	(0.075)	(0.070)
Amending	1.462***	(omitted)	0.799	1.457***	1.433***
	(0.071)		(0.759)	(0.079)	(0.073)
RCM Climate Salience	1.450**	1.081	1.166	1.488*	1.332
	(0.229)	(0.182)	(0.188)	(0.334)	(0.276)
Public Concern (CRP)	0.143***	0.012***	0.018***	0.092***	0.114***
	(0.023)	(0.002)	(0.003)	(0.017)	(0.020)
N	2,272	2,272	2,272	2,272	2,272

Notes: Hazard ratios (Cox proportional hazards). Standard errors in parentheses.

$^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Model 5A. *Variance Inflation Factor (VIF Test)*

Variable	**VIF**	**1/VIF**
Government Effectiveness	6.28	0.1592
Regulatory Quality	5.08	0.1969
GDP per Capita	2.73	0.3664
Political Stability	1.71	0.5852
Post Green Deal	1.29	0.7765
Amending	1.17	0.8519
Mean VIF	**3.04**	

Model 5A. *Effect of Administrative Quality and Political Stability (Green Deal Directives)*

	Pooled	**FE (CELEX)**	**RE (CELEX)**	**FE (Country)**	**RE (Country)**
Post Green Deal	1.029	0.035***	0.089***	1.730***	1.504***
	(0.051)	(0.009)	(0.013)	(0.122)	(0.094)
Amending	1.587***	(omitted)	1.124	1.526***	1.517***
	(0.077)		(0.926)	(0.081)	(0.077)
Government Effectiveness	0.994	1.189*	1.144	2.035***	1.747***
	(0.095)	(0.120)	(0.116)	(0.394)	(0.293)
Regulatory Quality	1.175	1.030	1.058	0.792	0.933
	(0.124)	(0.114)	(0.116)	(0.174)	(0.183)
GDP per Capita	1.000	0.999***	0.999**	0.999***	0.999***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Political Stability	0.897	1.135	1.187*	2.909***	2.521***
	(0.084)	(0.113)	(0.118)	(0.471)	(0.379)
N	2,325	2,325	2,325	2,325	2,325

Notes: Hazard ratios (Cox proportional hazards). Standard errors in parentheses.

$^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Appendix B. Regression outputs for Labour directives

Model 1B. *Variance Inflation Factor (VIF Test)*

Variable	**VIF**	**1/VIF**
Coalition Size	1.05	0.949
Subnat. Competence	1.05	0.949
Amending	1.02	0.979
Technical Cabinet	1.02	0.982
Mean VIF	**1.04**	

Model 1B. *Effect of Institutional Factors (Labour Directives)*

	Pooled	**FE (CELEX)**	**RE (CELEX)**	**FE (Country)**	**RE (Country)**
Amending	0.903	(omitted)	1.043	0.824**	0.830**
	(0.065)		(0.170)	(0.070)	(0.065)
Subnat. Competence	0.817***	0.791***	0.736***	1.494**	1.313*
	(0.062)	(0.070)	(0.062)	(0.283)	(0.211)
Coalition Size	0.938***	0.936***	0.933***	0.935	0.927*
	(0.023)	(0.023)	(0.023)	(0.043)	(0.036)
Technical Cabinet	1.558*	1.827**	1.657*	1.964**	1.667*
	(0.400)	(0.482)	(0.431)	(0.596)	(0.463)
N	1,023	1,023	1,023	1,023	1,023

Notes: Hazard ratios (Cox proportional hazards). Standard errors in parentheses.

$^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Model 2B. *Variance Inflation Factor (VIF Test)*

Variable	**VIF**	**1/VIF**
Left–Right	6.27	0.159
Liberty–Authority	3.58	0.280
State–Market	2.99	0.334
Pro–Anti EU	1.83	0.546
Technical Cabinet	1.01	0.991
Amending	1.01	0.995
Mean VIF	**2.78**	

Model 2B. *Effect of Political Orientation (Labour Directives)*

	Pooled	**FE (CELEX)**	**RE (CELEX)**	**FE (Country)**	**RE (Country)**
Amending	0.953	(omitted)	1.149	0.795**	0.806**
	(0.070)		(0.194)	(0.072)	(0.068)
Technical Cabinet	3.617**	7.966***	4.352***	7.608***	7.649***
	(1.843)	(4.343)	(2.242)	(5.810)	(4.458)
Left–Right	1.537***	1.619***	1.606***	1.221*	1.393***
	(0.113)	(0.126)	(0.123)	(0.128)	(0.134)
State–Market	0.681***	0.610***	0.653***	0.816**	0.702***
	(0.038)	(0.036)	(0.037)	(0.070)	(0.055)
Liberty–Authority	0.827***	0.856***	0.806***	0.739***	0.772***
	(0.039)	(0.044)	(0.041)	(0.057)	(0.051)
Pro–Anti EU	1.036	1.072*	1.023	0.703***	0.849***
	(0.036)	(0.041)	(0.037)	(0.059)	(0.053)
N	955	955	955	955	955

Notes: Hazard ratios (Cox proportional hazards). Standard errors in parentheses.

$^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Model 5B. *Variance Inflation Factor (VIF Test)*

Variable	**VIF**	**1/VIF**
Government Effectiveness	3.06	0.327
GDP per Capita	2.47	0.404
Political Stability	1.46	0.686
Amending	1.05	0.955
Technical Cabinet	1.03	0.973
Mean VIF	**1.81**	

Model 5B. *Governance Quality and Economic Capacity (Labour Directives)*

	Pooled	**FE (CELEX)**	**RE (CELEX)**	**FE (Country)**	**RE (Country)**
Amending	0.550***	(omitted)	0.434*	0.442***	0.486***
	(0.043)		(0.187)	(0.043)	(0.041)
Technical Cabinet	1.361	1.731**	1.611*	1.286	1.249
	(0.347)	(0.456)	(0.419)	(0.384)	(0.331)
Government Effectiveness	1.082	1.121	1.138	0.914	1.108
	(0.138)	(0.152)	(0.150)	(0.337)	(0.205)
Political Stability	0.687**	0.740*	0.717*	0.822	0.982
	(0.109)	(0.126)	(0.124)	(0.265)	(0.237)
GDP per Capita	1.000	1.000	1.000	0.999***	0.999
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
N	895	895	895	895	895

Notes: Hazard ratios (Cox proportional hazards). Standard errors in parentheses.

$^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Appendix C. Data sources

This appendix provides an overview of the main data sources used to construct the dataset employed in the empirical analysis. The Green Deal and labour directive data were first downloaded from EUR-Lex using a self-developed Python script and then merged with political, institutional, governance and socio-economic data from other sources. The integration and matching of datasets were conducted using Python, while data cleaning, transformation and survival analyses were performed with Stata.

EUR-Lex. The legal database of the European Union was used to extract all directives related to the Green Deal and the Labour policy field.

ParlGov Database (October 2023 version). Used to obtain detailed information on national governments, including cabinet composition, coalition size, ideology and duration.

Manifesto Project Dataset (MPDS2025a). Provided party-level ideological scores over environmental policies.

Worldwide Governance Indicators (World Bank). Provided measures of Government Effectiveness, Regulatory Quality, Rule of Law, Control of Corruption, Political Stability and Voice and Accountability.

Eurostat. Data on GDP per capita were taken from Eurostat.

Media Cloud and Google Trends. Used to construct annual indicators of media salience and public attention to climate change and the Green Deal.

Eurobarometer Surveys (GESIS). Used to obtain data on public opinion regarding climate change and environmental policy.

Appendix D. List of Directives Included in the Dataset

CELEX	Directive Name	Adoption	Deadline	Amending
32018L2001	Directive (EU) 2018/2001	11/12/2018	30/06/2021	No
32004L0101	Directive 2004/101/EC	27/10/2004	13/11/2005	Yes
32009L0029	Directive 2009/29/EC	23/04/2009	31/12/2012	Yes
32010L0031	Directive 2010/31/EU	19/05/2010	09/07/2012	No
32018L0410	Directive (EU) 2018/410	14/03/2018	09/10/2019	Yes
32018L0844	Directive (EU) 2018/844	30/05/2018	10/03/2020	Yes
32018L2002	Directive (EU) 2018/2002	11/12/2018	25/06/2020	Yes
32023L0958	Directive (EU) 2023/958	05/10/2023	31/12/2023	Yes
32023L0959	Directive (EU) 2023/959	05/10/2023	31/12/2023	Yes
32023L1791	Directive (EU) 2023/1791	13/09/2023	11/10/2025	No
32023L2413	Directive (EU) 2023/2413	20/11/2023	01/07/2024	Yes
32024L1275	Directive (EU) 2024/1275	24/04/2024	01/01/2025	No
32009L0028	Directive 2009/28/EC	23/04/2009	05/12/2010	No

Table D.1: EU Green Deal directives included in the dataset.

CELEX	Directive Name	Adoption	Deadline	Amending
32010L0018	Directive 2010/18/EU	07/07/2010	05/08/2012	Yes
32010L0041	Directive 2010/41/EU	16/04/2024	21/05/2016	No
32014L0054	Directive 2014/54/EU	16/04/2024	18/06/2016	No
32014L0067	Directive 2014/67/EU	15/05/2014	15/11/2016	Yes
32017L0159	Directive (EU) 2017/159	19/12/2016	15/11/2019	No
32018L0131	Directive (EU) 2018/131	15/02/2018	16/02/2020	Yes
32018L0957	Directive (EU) 2018/957	28/06/2018	30/07/2020	Yes
32019L0130	Directive (EU) 2019/130	16/01/2019	20/02/2021	Yes
32019L1152	Directive (EU) 2019/1152	20/06/2019	01/08/2022	No
32019L1158	Directive (EU) 2019/1158	20/06/2019	02/08/2022	No
32019L1937	Directive (EU) 2019/1937	23/10/2019	17/12/2021	No
32021L1883	Directive (EU) 2021/1883	20/10/2021	18/11/2023	No
32022L2041	Directive (EU) 2022/2041	19/10/2022	15/11/2024	No

Table D.2: EU labour directives included in the dataset.

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